Small Changes and Market Unraveling: the Role of Non-Exclusive Competition and Outside Options*

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Abstract

When do small deteriorations in market quality trigger sudden, widespread collapses in trade? Existing results show that markets are robust to small shocks under exclusive contracting, while collapse is possible under knife-edge conditions in non-exclusive contracting with linear preferences. We ask when collapse is a general phenomenon. In the non-exclusive environment of Attar et al. (2021), we first show markets are generically robust: remaining agents' marginal rates of substitution weakly increase after each exit, preserving trade. This robustness breaks down once outside options are generalized beyond autarky. We establish conditions for exit cascades: non-exclusive contracting, (partial) pooling, and outside options above the no-trade utility. A single exit then raises prices across all shared pooling layers, triggering a cascade that shuts down the entire market. The separating equilibria of Attar et al. (2014) are immune to such cascades even with generalized outside options due to the separation of contract pricing across types.

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1 Introduction

Financial crises are often sudden: credit markets that function smoothly for years can seize within weeks as risk accumulates and adverse selection worsens. As Krishnamurhty and Muir (2025) and Benmelech and Bergman (2018) document, small deteriorations in underlying fundamentals have historically preceded large, discontinuous collapses in market activity. Understanding the mechanisms that convert gradual risk accumulation into abrupt market shutdowns is therefore a central question in financial economics.

This paper asks: when can a small deterioration in the quality of market participants cause trade to collapse entirely? The canonical framework of Akerlof (1970) provides one answer under exclusive contracting—when there is a single market-clearing price, adverse selection can unravel trade entirely. But in many important markets—insurance, over-the-counter derivatives, collateralized debt and loan obligations—agents are free to trade simultaneously with multiple counterparties. This *non-exclusive contracting* environment, which better describes these markets, has a richer equilibrium structure, and it is far from obvious whether and when the Akerlof mechanism survives. We provide a complete characterization: market shutdowns require not only non-exclusive contracting but also outside options that are more valuable than autarky, and equilibrium that exhibits (partial) pooling across types. Without all three ingredients, markets are generically robust to small shocks. With all three, a single exit can cascade into full market collapse.

Two prior results are essential building blocks for understanding our contribution. Azevedo and Gottlieb (2017) develop a general model of perfectly competitive adverse selection under exclusive contracting that accommodates very general preferences, contracting spaces, and outside options. They show that their equilibrium is robust to small changes in fundamentals: outside of knife-edged cases, the "only way to produce large changes in equilibrium predictions is to considerably move average cost or demand curves." This establishes that exclusive contracting, by itself, cannot generate sudden market shutdowns—the mechanism must lie elsewhere. Attar et al. (2011) provide the first hint of where. In a non-exclusive version of Akerlof (1970) with linear preferences and capacity constraints, all trades collapse into a single pooling price, and this structure can generate knife-edged market unraveling: a small deterioration in fundamentals can lead an equilibrium with positive trade to one with none at all. However, the result is fragile and special—it requires linearity, a single pool, and the precise conditions under which the demand curve is tangent to the average cost curve.

Our main results complete this picture. First, we show that neither of these special features—exclusive contracting on one side, linear single-pool on the other—is what drives market fragility. In the general non-exclusive contracting environment of Attar et al. (2021), which features non-linear pricing and multiple types trading across many price layers, small cost increases *do not* generate exit cascades or market shutdowns. Even as the worst type exits and market prices deteriorate, all remaining agents' marginal rates of substitution weakly increase, preserving their incentives to trade. This extends the robustness intuition of Azevedo and Gottlieb (2017) to the non-exclusive contracting setting, and shows that non-exclusivity alone is insufficient. Second, once we generalize outside options—allowing agents to exit to utility levels strictly above autarky—small cost increases *can* cause a cascade of exits and complete market shutdown in Attar et al. (2021). The exit of the best type raises the equilibrium price in every shared pooling layer, lowering all remaining agents' total utility and potentially pushing them past their outside option threshold. One exit begets the next. Third, this cascade mechanism requires partial pooling in equilibrium. In the simultaneous contracting environment of Attar et al. (2014)—which admits only separating equilibria for non-zero trades—outside options cannot generate cascades even when generalized, because the exit of one type relaxes incentive compatibility for the remaining types, weakly improving their terms of trade. Fourth, economies with more *coarse* partitions of types are less prone to cascades: when adjacent types are pooled into a single category, the relative mass of each partition is larger, requiring a bigger shock to trigger the initial exit that cascades.

The key to understanding why the equilibrium structure of Attar et al. (2021) generates cascades—while Attar et al. (2011) only does so in a knife-edged way, and Attar et al. (2014)

not at all—lies in how types are linked through shared prices. In Attar et al. (2021), trades are organized as recursive layers along a convex-market tariff: all types share the first price layer, types 2 through n share the second, and so on. When one type exits, the cost of every shared layer rises for all remaining types. This *common price exposure* creates strategic complementarity in exit decisions that is absent from both the single-pool linear model of Attar et al. (2011) (where pooling is too coarse to differentiate types) and from separating environments. Outside options then provide the threshold that converts deteriorating terms of trade into an actual exit, resulting in additional discrete decline in terms of trade. In this sense, our results identify (partial) pooling with non-exclusivity and generalized outside options as the precise combination necessary for fragility—generalizing and sharpening the Attar et al. (2011) knife-edge insight.

The model requires no institutional details, dynamic mechanisms, or sentiment-driven runs: just a single-crossing property on preferences and a monotonicity condition on costs. This parsimony makes the framework widely applicable. CDO and CLO markets, interbank lending, insurance, and OTC derivative markets all feature non-exclusive contracting and partial pooling. Figures 1 and 2 illustrate the phenomenon directly: CDO/CLO issuance collapses discontinuously during crises even as underlying default rates increase only modestly, consistent with a cascade mechanism rather than a smooth Akerlof-style unraveling.

Finally, our results reveal a surprising dual role for outside options. In Attar et al. (2014), where equilibrium is separating, outside options can actually *open* markets: if one type's outside option is sufficiently attractive that it exits voluntarily, the adverse selection problem disappears and the remaining type can trade at its type-specific price—an endogenous screening device in the spirit of Board and Pycia (2014). It is only when partial pooling links agents' terms of trade that outside options flip from market-opening to market-closing.

The paper proceeds as follows. Section 2 examines two special cases that motivate our general analysis: the exclusive contracting environment of Azevedo and Gottlieb (2017), which is robust to small shocks, and the non-exclusive environment of Attar et al. (2011), which generates knife-edge unraveling under linear preferences and a single pooling price. Section 3 presents the general framework of Attar et al. (2021), including the convex market tariff, entry-proofness, and the layered equilibrium structure. Section 4 establishes that absent outside options, active markets are robust to small cost increases: as the worst type exits, remaining agents' marginal rates of substitution weakly increase, preserving trade. Section 5 introduces generalized outside options and derives necessary and sufficient conditions for a cascade of exits—partial pooling combined with outside options above autarky—with policy implications and microfoundations discussed throughout. Section 6 shows that the separating equilibria of Attar et al. (2014) are immune to cascades even with generalized outside options, isolating partial pooling as the essential ingredient for market fragility.

Relation to the Literature. Our paper sits at the intersection of two literatures: non-exclusive contracting under adverse selection, and the conditions under which adverse selection markets unravel.

Our paper relates to the literature studying non-exclusive contracting in economies with adverse selection pioneered by (e.g., Pauly, 1974; Jaynes, 1978; Hellwig, 1988; Glosten, 1994). Allocations in these non-exclusive contracting environments are recursive; agents trade in layers consisting of multiple contracts.[1] More recent results extend to generalized settings with divisible goods, general preferences, and multiple types (e.g., Attar et al., 2011, 2014, 2021; Dubey and Geanakoplos, 2019). Our paper builds directly upon Attar et al. (2021) and Dubey and Geanakoplos (2019), but with a different focus: we study the conditions under which small changes in the distribution of agents cause markets to unravel.

The closest predecessors within this literature are Attar et al. (2011), Attar et al. (2014), and Attar et al. (2021). Attar et al. (2011) show, in a non-exclusive version of Akerlof (1970) with linear preferences and capacity constraints, that markets can cease to function for high quality agents while remaining open for low quality agents.[2] Their framework generates a

[1] Bisin and Gottardi (1999, 2003) show that some form of non-linear pricing is needed to make compatible price taking behavior with asymmetric information.

[2] This is true aside from the trivial case where any buyer's expected valuation of the good is lower than the lower bound of the valuation of sellers.

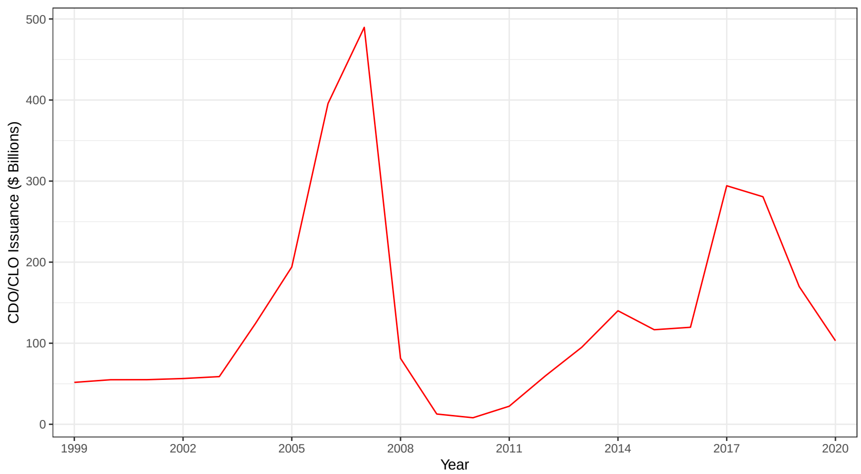

Figure 1: Issuance of Collateralized Debt Obligations and Collateralized Loan Obligations

Source: Securities Industry and Financial Markets Association.

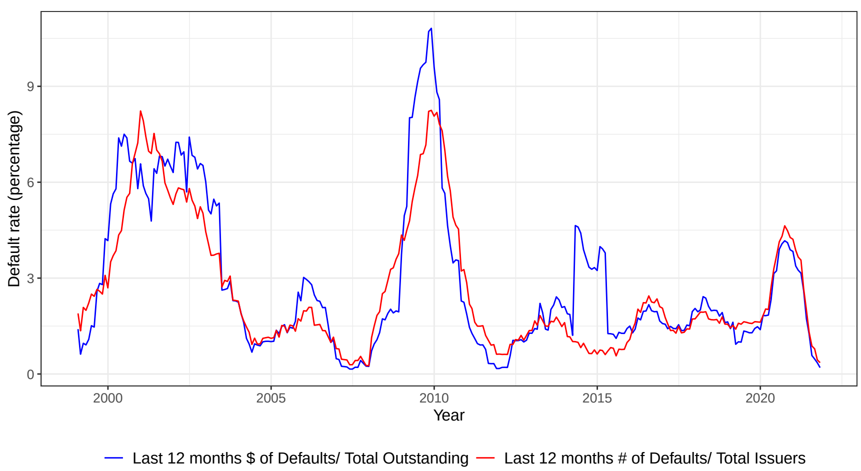

Figure 2: U.S. leveraged loan default rate

Note: CDO and CLO issuance plummets in times of crises, but the default rate of the underlying securities–leveraged loans–modestly increases.
Source: S&P Global Market Intelligence.

strategic foundation for an Akerlof-like single pooling equilibrium, and market unraveling is possible at a knife-edge: a small deterioration in fundamentals can move equilibrium from positive trade to none. However, as we show, this result is special to the linear, single-pool structure; it does not generalize once preferences are non-linear and multiple price layers emerge. Attar et al. (2014) and Attar et al. (2021) consider general preferences and non-linear pricing, but do not establish a market unraveling result—indeed, a deterioration in the type distribution can increase the likelihood of positive trades by bad types rather than triggering a shutdown. We fill this gap: we derive the general conditions under which small changes in fundamentals cause markets subject to adverse selection with non-exclusive contracting to unravel and shut down. Our contribution relative to Attar et al. (2011) is to move from a knife-edge special case to a generalized condition that applies across general preferences and multi-layered equilibria.

Our focus on the unraveling of competitive equilibrium due to adverse selection dates back to Akerlof (1970) and Rothschild and Stiglitz (1976), and has recently been generalized by Hendren (2013, 2014) and Azevedo and Gottlieb (2017). Azevedo and Gottlieb (2017) develop a general model of perfectly competitive adverse selection under exclusive contracting that accommodates general preferences, contracting spaces, and outside options. Though robustness is not their main focus, they establish that equilibrium is resilient to small changes in fundamentals: complete market unraveling outside of knife-edged cases requires large shocks to average cost or demand. Our paper is the non-exclusive contracting counterpart to this result. We first confirm that their robustness finding extends to non-exclusive environments—absent outside options, active markets do not unravel after small cost increases. We then identify the precise conditions under which robustness *fails*: the combination of (partial) pooling in equilibrium and generalized outside options. This provides a sharp characterization that is absent from Azevedo and Gottlieb (2017) because their exclusive contracting environment cannot feature the shared-price cross-subsidization that creates strategic complementarity in exit decisions. More recently, Kurlat (2013) embeds the Akerlof insight into macro models to explain how information frictions amplify business cycles, and Malherbe (2014) and Asriyan et al. (2019) show how long-term asset markets can suffer self-fulfilling liquidity freezes. Unlike these papers, our mechanism requires neither dynamics nor belief-driven runs; it operates in a static, belief-free framework through the equilibrium structure alone.

Hendren (2014) shows that equilibrium in insurance economies with exclusive contracting must unravel either via Akerlof-pricing or failure to satisfy competitive-Nash equilibrium à la Rothschild-Stiglitz when the type distribution contains a continuous interval or a type with accident probability equal to 1. We show that this result is sensitive to the exclusive contracting assumption: under non-exclusive contracting, entry of a type with accident probability equal to 1 is not sufficient to cause markets to unravel. Levy and Veiga (2021) show that equilibrium in Azevedo and Gottlieb (2017) may not exist if the cost of supplying agents is unbounded. We show that with non-exclusive contracting, both Rothschild-Stiglitz and Akerlof unraveling are possible even with bounded cost, and that agents' entry and exit decisions can be strategic complements or substitutes depending on the structure of outside options. Kessler (1998) and Levin (2001) show similar effects arising from changes in the information structure of the economy.

Recent applications of non-exclusive contracting under adverse selection include security design (e.g. Asriyan and Vanasco, 2024) and asset markets with heterogeneously informed buyers (e.g. Kurlat, 2016). A common feature of these environments is that equilibrium is never fully separating; there is always some form of cross-subsidization. Our model inherits this semi-pooling structure, but our focus is on how it interacts with outside options to generate market fragility. Notably, Asriyan and Vanasco (2024) and Bigio and Shi (2025) show that partial pooling is the *unique* equilibrium outcome in security design and repurchase option markets, respectively—precisely the environments our results identify as vulnerable to exit cascades. Auster et al. (2021) study a form of non-exclusivity in search with adverse selection, where fully separating equilibria are precluded because high types hedge against unemployment by applying to low-wage firms.

Philippon and Skreta (2012) show that the failure of the price mechanism and market unraveling justify public interventions during liquidity or credit freezes. A key insight in their framework is that interventions affect which agents choose to participate in government programs, which in turn impacts market trade. In our non-exclusive contracting framework, policies that increase entry cost prevent market unraveling only if they can discriminate among types; a uniform cost increase makes markets more prone to unraveling because the best types exit first, raising the cost of trade for all remaining types.

2 Special Cases with Market Shutdowns

Before developing our formal analysis, it is useful to understand how market unraveling arises—and fails to arise—in existing environments. We begin with Azevedo and Gottlieb (2017), who develop a general model of perfectly competitive adverse selection under exclusive contracting. Their framework is highly flexible: preferences and the contracting space are general, equilibrium can be separating, pooling, or partial pooling, and outside options are permitted. Though robustness is not their main focus, they establish in Section 4.3 that equilibrium is resistant to small changes in fundamentals—"the only way to produce large changes in equilibrium predictions is to considerably move average cost or demand curves." Thus, exit cascades and complete market unraveling are not generic features of exclusive contracting environments.[3]

This motivates the study of non-exclusive contracting environments, where agents trade simultaneously with multiple counterparties. We first consider Attar et al. (2011), who establish a general non-exclusive contracting framework with linear preferences and capacity-constrained agents. These assumptions generate a strategic foundation for an Akerlof (1970)-like equilibrium: all trades collapse into a single pool at a price equal to the average quality of the good, and agents either trade their entire endowment or nothing at all. Within this single-pool structure, market unraveling is possible at a knife-edge: when equilibrium gains from trade are near zero, a small deterioration in fundamentals can move the market from positive trade to none.[4]

[3] Market unraveling after a small shock in the spirit of Akerlof (1970) is possible, but requires the demand curve and the average cost curve to be tangent—a knife-edge condition. See the Supplementary Material to Azevedo and Gottlieb (2017) for details.

[4] This knife-edge property is confirmed by Proposition 5 of Attar et al. (2011), which shows that whenever the good is traded in strictly positive quantity, equilibrium quantities remain positive after further exits. Complete unraveling therefore requires the market to be at the margin of viability to begin with.

Table 1: Conditions for Market Collapse from Small Cost Increases

Framework	Non-Excl. Contract.	Pooling in Equil.	Outside Opt. > Autarky	Cascade Possible?
Azevedo & Gottlieb (2017)	No	Varies	Yes	No*
Attar et al. (2011)	Yes	Full	No	Knife-edge*
Attar et al. (2014)	Yes	No	No	No
Attar et al. (2014) + Outside Option	**Yes**	**No**	**Yes**	**No**
Attar et al. (2021)	Yes	Partial	No	No
Attar et al. (2021) + Outside Option	**Yes**	**Partial**	**Yes**	**YES**

*Requires knife-edge conditions or large shocks.

Whether such unraveling is possible beyond this special case—with general preferences, non-linear pricing, and multi-layered equilibria—is the central question of our paper. We consider the two canonical generalizations: the simultaneous contracting environment of Attar et al. (2014) and the sequential contracting environment of Attar et al. (2021). We show that neither environment generates exit cascades from small cost increases alone. Once outside options are generalized, however, the environments diverge sharply: cascades become possible in Attar et al. (2021) but remain impossible in Attar et al. (2014). The difference, as we show, traces entirely to equilibrium structure—partial pooling in Attar et al. (2021) creates strategic complementarity in exit decisions that is absent from the separating equilibria of Attar et al. (2014). Table 1 summarizes the main message of this paper; exit cascades outside of knife-edge cases are only possible when three features are met: non-exclusive contracting, (partial) pooling in equilibrium, and outside options exceeding the autarky payoff.

3 Model of Attar et al. (2021)

We present the model of Attar et al. (2021). The demand side of the market consists of a finite number of privately informed agents indexed by $i \in I \equiv \{1, .., n\}$ with a strictly positive measure of each type, m_i. Utility for each type is given by $u_i(q, t)$ and assumed to be continuous, quasi-concave in the arguments and strictly decreasing in t. Here q represents the quantity of a good consumed and t is the transfer required to obtain it. An important feature of the model is that privately informed types are ordered by a single-crossing property (Milgrom and Shannon, 1994), $\forall i < j,\ q < q',\ t,\ t'$:

$$u_i(q,t) \leq u_i(q',t') \Rightarrow u_j(q,t) < u_j(q',t').$$

Single-crossing implies that a higher type is at least as willing as a lower type to trade an additional unit of the good for an additional transfer.[5] Since utility functions need not be differentiable, we define the marginal rate of substitution without imposing differentiability. Let $\tau_i(q,t)$ be the supremum set of prices, p, such that

$$\tau_i(q,t) \equiv \sup \left\{ p : u_i(q,t) < \max_{q' \geq 0} u_i(q+q', t+pq') \right\}.$$

Hence, $\tau_i(q,t)$ is the marginal rate of substitution for agent i at consumption bundle (q,t)—the supremum price per unit at which agent i is willing to trade an additional quantity $q' > 0$ on top of (q,t). An important assumption for our analysis is that, absent a transfer, a strictly positive endowment of q lowers agents' marginal rate of substitution:

Assumption 1 $\tau_i(q,0) \leq \tau_i(0,0)$, $\forall i, q > 0$.[6]

By way of concrete examples, the model translates into the insurance economy of Rothschild and Stiglitz (1976), where i indicates an agent's probability of loss, q is the amount of insurance purchased, and t is the insurance premium. In a credit economy, i indexes borrower default probability, q is the loan quantity demanded, and t is the gross loan promise made to the lender.

Contracts, defined by the pair (q,t) for $q \geq 0$, are supplied by competitive risk-neutral, expected profit maximizers. The suppliers possess a linear production technology with a unit cost, $c_i > 0$, which is increasing in i. Adverse selection arises because c_i is increasing in type:

[5] Here we are using the strict single-crossing condition rather than the weak version, which allows for equality. This is because we focus on the strict notion of market breakdown in light of Corollary 1 in Attar et al. (2021).

[6] The same assumption is also used in Attar et al. (2021) in their analysis.

higher types wish to trade more but are more costly to serve. Define by $\bar{c}_i$ the expected unit cost of serving $j \geq i$ (the upper-tail conditional expected cost) given that higher types will be willing to trade any contract offered to type i. Formally,

$$\bar{c}_i \equiv E[c_j | j \geq i] = \frac{\sum\limits_{j \geq i} m_j c_j}{\sum\limits_{j \geq i} m_j}. \tag{1}$$

3.1 Equilibrium

We start with an equilibrium in which the market is inactive: there is no set of contracts an entrant can propose that leads to positive trades with nonnegative profits. Such a market is "entry-proof" in the sense of Attar et al. (2021).

The key is to first consider the no-trade contract, $(0, 0)$, as any agent's outside option. (Our subsequent analysis generalizes the outside option utility to be greater than that of the no-trade contract.) Then, a market is entry proof iff, for any menu of contracts an entrant offers, the buyer's best response earns the entrant zero expected profit. That is, the entry-proof condition is given by

$$\textbf{Condition EP}: \qquad \tau_i(0, 0) \leq \bar{c}_i \qquad \forall i. \tag{2}$$

Condition EP simply says that there will be no trade in a market when the cost of offering a contract for agent i exceeds type i's marginal utility of not trading, given that all other types higher than i must also be served.[7] Theorem 1 in Attar et al. (2021) states that **Condition EP** is necessary and sufficient for markets to be inactive.

Now we consider an equilibrium in which the market is active, so there is positive quantity of trades. Trade in the market is non-exclusive in the sense that no agent can be stricken from trading with multiple firms or suppliers. Therefore, we must define the market tariff,

[7]We use contracts and markets interchangeably, treating all contracts with potentially different price-quantity pairs as individual markets following Bisin and Gottardi (1999); Dubey and Geanakoplos (2002); Dubey et al. (2005).

or the minimum aggregate transfer that is made across active markets to obtain aggregate consumption, q. Define the aggregate market tariff by $T(q)$. We will assume that $T(q)$ is convex and the domain is a compact interval with lower bound equal to 0.[8] Then, all agents choose q_i to maximize $u_i(q_i, T(q_i))$. An allocation, $(q_i, T(q_i))_{i \in I}$ is *implemented* by the market tariff, T, if $q_i = \arg\max_q u_i(q, T(q))$. This allocation is *budget feasible* if suppliers make non-negative expected profits at the market tariff

$$\sum_i m_i \left[T(q_i) - c_i q_i\right] \geq 0. \tag{3}$$

Under an exclusive contracting environment, an entrant only needs to consider the agent's direct utility functions. However, under a non-exclusive contracting environment, an agent can combine any menu of potential new contracts with trades along the existing market tariff, T. Thus, an active market is entry proof if an entrant cannot make positive expected profits given agents' ability to combine contracts. Therefore, the entrant faces types with indirect utility functions of trading a proposed new contract (q', t') in addition to the existing allocation $(q, T(q))$:

$$u_i^T(q', t') \equiv \max \left\{u_i(q + q', T(q) + t' : q)\right\} \tag{4}$$

From the entrant's perspective, an agent's individual rationality constraint is determined by the indirect utility of not trading the proposed contract on top of the existing market tariff, $u_i^T(0, 0)$. We can define the marginal rate of substitution along the indirect utility functions as above by $\tau_i^T(q', t')$.[9] As shown by Attar et al. (2021), the indirect utility functions also satisfy single-crossing, since the primitives do.

Attar et al. (2021) impose an additional assumption, slightly stronger than Assumption

[8] See Section 5.2.3 for more details on strategic foundations of $T(q)$.

[9] This is possible because: (i) the maximizers in (4) are continuous by Berge's Maximization Theorem; (ii) the market tariff T is convex; (iii) the utility functions $u_i(q, t)$ are weakly quasi-concave in (q, t) and strictly decreasing in t; and (iv) therefore the indirect utility functions u_i^T are weakly quasi-concave in (q, t) and strictly decreasing in t.

1, requiring that the marginal rate of substitution is nonincreasing in q for any type i and transfer t:

Assumption 2 *For all i and t, $\tau_i(q,t)$ is nonincreasing in q.*

Intuitively, this assumption implies that a higher quantity always reduces each agent's willingness to pay for any additional quantity. For the given market tariff, T, and the allocation, $(q_i, T(q_i))$, we can define $\tau_i^T(0,0)$ as the supremum of the set of prices p for the indirect utility function, $u_i^T(0,0)$—that is,

$$u_i(q_i, T(q_i)) = u_i^T(0,0) < \max\left\{u_i^T(q', pq') : q'\right\} = \max\left\{u_i(q+q', T(q)+pq' : q, q')\right\}.$$

With this, we can invoke the necessity and sufficiency result of **Condition EP** to state that a market tariff is entry-proof iff:

$$\forall\, i, \quad \tau_i^T(0,0) \leq \bar{c}_i. \tag{5}$$

This condition states that an active market is entry-proof if and only if the cost required to enter the market exceeds the willingness of each agent to trade the contract on top of the allocation they may already obtain. For each agent, the utility they receive from their market trades must be at least as large as trading the proposed new contract given the cost required to serve the market:

$$\text{for each } i, \quad u_i(q_i, T(q_i)) \geq \max\left\{u_i(q+q', T(q)+\bar{c}_i q' : q, q')\right\}. \tag{6}$$

From the entrant's perspective, by convention of letting $q_0 \equiv 0$, setting $q' = q_i - q$, and applying condition (6) for each layer, $q \in [q_{i-1}, q_i]$, we see that the implied market tariff necessary to induce entry must be at least as large as the pre-entry tariff:

$$\text{for each } i \text{ and } q \in [q_{i-1}, q_i], \quad T(q_i) \leq T(q) + \bar{c}_i(q_i - q). \tag{7}$$

For a given $q = q_{i-1}$, we have $T(q_i) \leq T(q_{i-1}) + \bar{c}_i(q_i - q_{i-1})$. Using the budget-feasibility condition from (3), and re-writing it in terms of layers, we have

$$\sum_i \left(\sum_{j \geq i} m_j \right) [T(q_i) - T(q_{i-1}) - \bar{c}_i(q_i - q_{i-1})] \geq 0. \tag{8}$$

Therefore, combining (7) and (8) implies

$$T(q_i) = T(q_{i-1}) + \bar{c}_i(q_i - q_{i-1}). \tag{9}$$

It must also be true that the allocation, $u_i(q_i, T(q_i))$, implied by the new layer, $q_i - q_{i-1}$ maximizes the utility of all the agents that choose it, given that the new market tariff must rise to serve all agents. Hence, for each i,

$$u_i(q_i, T(q_i)) = \max \left\{ u_i(q_{i-1} + q', T(q_{i-1}) + \bar{c}_i q' : q') \right\}. \tag{10}$$

Finally, because the tariff is convex and satisfies (7) and (10), it must be affine with slope $\bar{c}_i$ over the interval $[q_{i-1}, q_i]$.

With this, Attar et al. (2021) state **Theorem 2**—An allocation $(q_i, T(q_i))_{i \in I}$ is budget-feasible and implemented by an entry-proof convex market tariff, T, with domain $[0, q_n]$ if and only if

1. $(q_0, T(q_0)) \equiv (0, 0)$,

2. $q_i - q_{i-1} \in \arg\max \left\{ u_i(q_{i-1} + q', T(q_{i-1}) + \bar{c}_i q' : q') \right\}$ for each i,

3. $q_{i-1} < q_i \Rightarrow T$ is affine with slope $\bar{c}_i$ over $[q_i, q_{i-1}]$ for each i.

To sum up, an affine convex market tariff is entry proof as long as the upper-tail conditional cost of entry exceeds the marginal willingness of all agents to trade the additional layer on top of their pre-entry allocation. This market tariff consists of layers of trade with unit prices $\bar{c}_i$ that trace out a polygon with an upward kink at each $q_{i+1} \geq q_i$ for each $i \in I$.

We take the strategic foundations of the allocations characterized in **Theorem 2** above as given. There are two ways to micro-found this allocation. The first is the discriminatory ascending-price auction used in Attar et al. (2021). In particular, each time a new price is quoted, each seller publicly announces the maximum quantity he stands ready to trade at this price. Once this auctioning phase is completed, the buyer decides which quantities to purchase from which sellers in a non-exclusive way. The second is the notion of competitive-pooling in general equilibrium pioneered by Dubey and Geanakoplos (2002) and extended to a non-exclusive contracting by Dubey and Geanakoplos (2019).

4 Cost Increases and Market Shutdowns

We now characterize how the equilibrium allocation responds when the cost of serving the market increases. Such cost increases may reflect a deterioration in the quality of the agent pool—for example, a surge in borrower defaults—or any other shock that raises the unit cost c_i. We define a *market shutdown* as the comparative static in which a previously active market becomes inactive: trade falls to zero.

To fix ideas, recall that the unit cost to supply a contract for each type is c_i. Now suppose that the cost to serve types increase to $\tilde{c}_i = c_i + \epsilon_i$, $\epsilon_i \geq 0$, $\forall i \in I$. The resulting upper-tail conditional expected cost of serving all agents must rise as long as there exists at least some type with $\epsilon_i > 0$. Define the new upper-tail conditional expected cost for any agent i by

$$\tilde{\bar{c}}_i \equiv \frac{\sum_{j \geq i} m_j \tilde{c}_j}{\sum_{j \geq i} m_j}, \tag{11}$$

and $\tilde{\bar{c}}_i \geq \bar{c}_i$. Then, it is clear that any market that was inactive *ex ante* the cost increase remains inactive *ex post* because **Condition EP** implies $\tau_i(0,0) \leq \bar{c}_i \leq \tilde{\bar{c}}_i$. Intuitively, making the average quality of the pool worse will never lead to the opening of new markets.

When do cost increases lead any active markets to become inactive and shut down? An active market satisfies $\tau_i(0,0) > \bar{c}_i$ for some i. If a market was active and $\tau_i(0,0) > \tilde{\bar{c}}_i$ and

$\tilde{q}_i > \tilde{q}_{i-1}$ for some i, it will remain active at a new equilibrium level of trade $(\tilde{q}_i)_{i\in I}$. The cost increase from c_i to $\tilde{c}_i$ causes the slope of the market tariff to weakly rise along all segments of the polygon tracing out the new market tariff, $\tilde{T}(\tilde{q})$. Active markets shut down only when $\tau_i^{\tilde{T}}(0,0) \leq \tilde{\bar{c}}_i$ for all i, which by (11) requires sufficiently large cost increases. Formally:

Theorem 1 (Cost Increases and Market Shutdowns)

Suppose c_i weakly increases to $\tilde{c}_i$ for all $i \in I$, where $c_j < \tilde{c}_j$ for some $j \in I$. Then, for the new equilibrium affine aggregate tariff, $\tilde{T}$, and new quantities, $(\tilde{q}_i)_{i\in I}$, the following hold:

1. *All inactive markets (market shutdown) remain inactive under the new equilibrium.*

2. *All active markets will remain active and have the new steeper equilibrium slope for the aggregate tariff, $\tilde{T}$, as $\tilde{\bar{c}}_i \geq \bar{c}_i$ over $[\tilde{q}_{i-1}, \tilde{q}_i]$ for every i, if $\tilde{q}_i > \tilde{q}_{i-1}$.*

3. *Active markets become inactive and shut down iff $\tau_i^{\tilde{T}}(0,0) \leq \tilde{\bar{c}}_i$ and $\tilde{q}_i = \tilde{q}_{i-1}$ for each i.*

Proof. Statements 1 and 3 are trivial as explained in the discussion before the theorem. We show that statement 2 holds.

First, consider type 1. Suppose that agent 1 was trading a positive quantity $q_1 > 0$, without loss of generality. Suppose that type 1 agent exits the market after the cost increase. Then, the quantity demanded trivially decreases as $\tilde{q}_1 = 0 < q_1$, and the market tariff increases as $\tilde{T}(q) \geq \tilde{\bar{c}}_2 q > \bar{c}_2 q > \bar{c}_1 q$ for any q in $[0, q_1]$ and $[q_1, q_2]$, regardless of the new trading quantities $\tilde{q}_2, \tilde{q}_3, \ldots, \tilde{q}_n$. Suppose that type 1 agent still trades a positive quantity in the market. The market tariff trivially increases to $\tilde{T}(q) = \tilde{\bar{c}}_1 q$ for any q in $[0, \tilde{q}_1]$ as in the previous case. Given the new optimal quantity for type 1, $\tilde{q}_1$, type 2 agent will face a higher slope of the affine tariff for the quantity $\tilde{q}_2 \geq \tilde{q}_1$ as

$$\frac{\tilde{T}(\tilde{q}_2) - \tilde{T}(\tilde{q}_1)}{\tilde{q}_2 - \tilde{q}_1} \geq \frac{\tilde{\bar{c}}_1\tilde{q}_1 + \tilde{\bar{c}}_2(\tilde{q}_2 - \tilde{q}_1) - \tilde{\bar{c}}_1\tilde{q}_1}{\tilde{q}_2 - \tilde{q}_1} = \tilde{\bar{c}}_2 \geq \bar{c}_2 = \frac{T(q_2) - T(q_1)}{q_2 - q_1},$$

because $\tilde{\bar{c}}_j \geq \bar{c}_j$ for any j.

Now consider an arbitrary type $i > 2$. Following the previous argument results in

$$\frac{\tilde{T}(\tilde{q}_i) - \tilde{T}(\tilde{q}_{i-1})}{\tilde{q}_i - \tilde{q}_{i-1}} \geq \frac{\sum_{j<i} \tilde{\bar{c}}_j \left(\tilde{q}_j - \tilde{q}_{j-1}\right) + \tilde{\bar{c}}_i(\tilde{q}_i - \tilde{q}_{i-1}) - \sum_{j<i} \tilde{\bar{c}}_j \left(\tilde{q}_j - \tilde{q}_{j-1}\right)}{\tilde{q}_i - \tilde{q}_{i-1}} = \tilde{\bar{c}}_i$$
$$\geq \bar{c}_i = \frac{T(q_i) - T(q_{i-1})}{q_i - q_{i-1}},$$

with the convention $\tilde{q}_0 \equiv 0$. Thus, the new slope for the affine market tariff $\tilde{T}$ becomes steeper in each and every interval of the new optimal quantity layers. ■

Theorem 1 states general properties of active and inactive markets under the new equilibrium. One important implication is that active markets will completely shut down only if $\tau_i(0,0) \leq \tilde{\bar{c}}_i$ for all i. Hence, the cost increase must be sufficiently large to generate market shutdowns.[10] Our market shutdown result extends the Akerlof-like unraveling result established in the non-exclusive contracting setting of Attar et al. (2011) to more general preferences and non-linear pricing. Their paper shows that even in the strategic setting of Rothschild and Stiglitz (1976), equilibrium always exists and is unique with non-exclusive competition. Hence markets fail in the Akerlof sense when the unique equilibrium is the no-trade equilibrium.

Moreover, Theorem 1 implies that no trade occurs in non-exclusive settings even when the separating equilibrium of Rothschild and Stiglitz (1976) exists under an exclusive contracting environment. For example, consider the 2-agent case. Under the exclusive contracting environment a pure-strategy equilibrium exists as a separating equilibrium as long as the payoff from a separating contract dominates the payoff from a pooling contract as

$$u_1(q_{RS}, c_1 q_{RS}) \geq \max_q u_1(q, \bar{c}_1 q),$$

[10] Azevedo and Gottlieb (2017) establish a similar result in the exclusive contracting environment; we extend their finding to the non-exclusive contracting case.

where q_{RS} is the quantity that satisfies

$$\max_q u_2(q, c_2 q) = u_2(q_{RS}, c_1 q_{RS}).$$

Under this simplified setup, the market shutdown condition under the non-exclusive contracting environment is

$$\begin{aligned} &\tau_1(0,0) \leq \bar{c}_1 \\ \Rightarrow\ &u_1(0,0) \geq \max_q u_1(q, \bar{c}_1 q) \end{aligned}$$

Combining the equilibrium existence condition for Rothschild and Stiglitz (1976) and our entry-proof market shutdown condition implies

$$u_1(q_{RS}, c_1 q_{RS}) \geq u_1(0,0) \geq \max_q u_1(q, \bar{c}_1 q),$$

hence, introducing non-exclusive contracting can generate market shutdowns for the same parameter values that guarantee existence of a separating equilibrium under the exclusive contracting of Rothschild and Stiglitz (1976). The intuition is as follows. Under non-exclusive contracting, full separation is not attainable and equilibrium always contains partial pooling. The market shuts down when the no-trade contract dominates the partial pooling contract for agent 1, which occurs when the mass of bad types is sufficiently large. Under exclusive contracting, by contrast, the separating equilibrium exists precisely when the mass of bad types is large enough to make the pooling contract unattractive relative to full separation. Since no trade under non-exclusive contracting requires a larger mass of bad types than what makes pooling unattractive under exclusive contracting, market shutdown under non-exclusive contracting implies existence of a separating equilibrium under exclusive contracting.

Our next result addresses which layers are most susceptible to agents exiting. The single-crossing condition implies that higher types have higher willingness to trade at higher prices. Hence, as costs increase the market tariff along each layer, the lowest types exit the market first. This is easiest to see for the case where the first type $i = 1$ is just indifferent to trading given the upper-tail conditional cost of serving all greater types $j > 1$ where $\tau_i(0,0) = \bar{c}_i < \tilde{\bar{c}}_i$, while other agents $j > 1$ could still have $\tau_j^{\tilde{T}}(0,0) > \tilde{\bar{c}}_j$. In this case, type 1 is no longer willing to trade at the new market tariff, $\tilde{T}(\tilde{q})$, given the additional cost required to make the tariff budget-feasible while all types $j > 1$ remain active in the market. Hence, only the first layer of trade, q_1, becomes inactive.

Proposition 1 *In equilibrium, the lowest type (cost) agents exit first as costs increase.*

Proof. First, the marginal rate of substitution, $\tau_i(q,t)$, is nonincreasing in q. Second, utility, $u_i(q,t)$, is continuous and decreasing in t. Combining this with strict single-crossing, we have $u_{i-1}(0,0) \leq u_{i-1}(\tilde{q}_{i-1}, \tilde{T}(\tilde{q}_{i-1})) \Rightarrow u_i(0,0) < u_i(\tilde{q}_{i-1}, \tilde{T}(\tilde{q}_{i-1}))$. By continuity, there exists $(\tilde{c}_j)_{j \in I}$ such that $\tilde{T}$ makes $u_{i-1}(0,0) > \max_q u_{i-1}(q, \tilde{T}(q))$ and $u_i(0,0) < \max_q u_i(q, \tilde{T}(q))$. Again by single-crossing, if $i-1$ exits, then any type $j < i-1$ also exits. ■

An important implication of Proposition 1 is that *small* cost increases have only marginal negative effects on active markets. The reason is that when the first type drops out and the first layer of trade, $(q_1, \tilde{T}(q_1))$, is removed from the market, the marginal value of each remaining layer for each agent goes up relative to the alternative of no trade at $(0,0)$. In other words, $\tau_i(0, \tilde{T}(0)) \geq \tau_i(q_1, \tilde{T}(q_1))$. Hence, the spillovers from agents exiting *increase* rather than decrease the incentives for all remaining agents to trade. This force keeps the remaining markets active. Therefore, **Condition EP** is robust in the sense that complete market unraveling under non-exclusive contracting requires a sufficiently large cost increase, consistent with Azevedo and Gottlieb (2017).

5 Outside Options and Market Shutdown

We now show that a cascade of exits—a sequential shutdown of multiple markets—is possible upon small increases in costs when outside options are present. An outside option represents an alternative to market participation: it may reflect a contractual barrier, a cost of entry such as the time and effort required to sign contracts or search for suppliers, or access to an external market that offers a reservation utility. The key feature is that agents compare the value of their outside option to the utility they obtain from market trades.

Denote the utility from outside options (and not entering the market) as γ_i for type i. Type i now compares the set of market contracts to their outside option. Thus, the individual rationality condition for i becomes

$$V_i(T) = \max \left\{ \gamma_i, \max_{q \geq 0} u_i(q, T(q)) \right\} \tag{12}$$

for a given aggregate market tariff T.[11]

5.1 Cascade of Exits

In this subsection, we present our main result: a sequential market shutdown arising from a cascade of exits, in which the exit of one agent triggers another's. The mechanism rests on the spillover effects of exits across agents in the presence of outside options. These spillover effects are amplified through partial pooling, because each agent's exit generates a lump-sum decline in utility for all other agents participating in the same pool. If the initial exit triggers subsequent exits, the impact of each additional exit accumulates, generating amplification through multiple exits. We show that the spillovers needed to generate a market shutdown do not exist in the model without outside options.

[11] Note outside options are irrelevant for equilibrium when they provide utility less than no trade.

We assume the following condition on the distribution of outside option values:

$$u_i(q,t) \geq \gamma_i \Rightarrow u_j(q,t) > \gamma_j \qquad \forall i < j,\ \forall q > 0,\ \forall t. \tag{13}$$

This condition implies that if agent i prefers a contract (q,t) with a positive quantity to the outside option, then agent $j > i$ also prefers the contract to the outside option, which is in line with the idea of single-crossing property. We discuss the role and micro-foundation of this assumption in subsection 5.2.3.[12]

Since agents who enter the market optimize against the aggregate market tariff just as in the baseline model, the characterization in Theorem 2 of Attar et al. (2021) for budget-feasible, entry-proof market tariffs carries over directly. The aggregate entry-proof convex market tariff T has slope $\bar{c}_i$ over each layer $[q_{i-1}, q_i]$, with the convention $q_0 = 0$. Accordingly, Statements 1 and 2 of Theorem 1 and Proposition 1 continue to hold, giving us the following.

Proposition 2 *In any active market equilibrium, there is a cutoff-type $\theta \in I \cup \{0\}$ such that any agents with type less than or equal to θ exit the market and any agents with type greater than θ remain in the market.*

Proposition 2 follows from Proposition 1, as the lowest type agents exit first in any active market equilibrium. Proposition 2 will be important for eliminating any possible nonexistence of equilibrium discussed in the following subsection B.1.

We now use Proposition 2 to discuss how exits of the lowest types can generate a further cascade. Agents decide to enter or remain in the market by comparing their total utility from market trades with the value of their outside option, so they exit whenever the outside option is preferable. Because the lowest types exit first, all remaining agents face a higher aggregate market tariff, which lowers their total utility. Unlike the baseline model without outside options—where remaining agents only need their marginal rate of substitution at no-trade to exceed the market cost, as in (2)—agents with outside options compare their

[12] See Appendix B.1 for a discussion on the impact of this assumption on equilibrium existence.

total utility from market trades against γ_i. This comparison can trigger a cascade: even agents who would have stayed under the baseline may exit once their total utility falls below their outside option threshold.

We now formalize the cascade argument. Suppose that agent 1 exits the market after costs increase, which occurs when **Condition IT** holds:

$$\max_{q\geq 0} u_1(q, \tilde{\tilde{c}}_1 q) \leq \gamma_1. \tag{14}$$

Note that we do not need to check **Condition EP**, $\tau_1(0,0) \leq \tilde{\tilde{c}}_1$, because each agent prefers the outside option over the no-trade contract, i.e., $\gamma_i \geq u_i(0,0)$ for any i. We can extend this intuitive condition to a condition that any agent with type j that is below i exits the market. We denote such a condition, **Condition MS**(i), as:

$$\max_{q\geq 0} u_j\left(q, \tilde{\tilde{c}}_j q\right) \leq \gamma_j, \qquad \forall j < i, \tag{15}$$

where agents up to type i exit the market. **Condition MS**(i) is stricter than **Condition EP** because outside options introduce an additional reason to exit beyond the MRS condition alone. Agents who would remain active under **Condition EP** may nonetheless exit when their total utility from trade falls below their outside option. We formally show that **Condition MS**(i) is necessary and sufficient to generate a cascade of exits up to agent i.

Theorem 2 (Cascade of Exits) *Any agent $j < i$ exits the market in equilibrium iff **Condition MS**(i) is satisfied.*

Proof. The proof of sufficiency is straightforward as **Condition MS**(i) prevents entry of agents with type $j < i$ for the entry-proof market tariffs.

Now consider the proof of necessity. By Proposition 2, we check the lowest agent's entry decision for each candidate equilibrium. Consider an equilibrium in which agent 1 also enters the market. Then, the corresponding aggregate market tariff will be

$$T^0(q) \equiv \sum_{i \in I} \tilde{\tilde{c}}_i (q - q^0_{i-1}) \mathbf{1} \left\{ q \in [q^0_{i-1}, q^0_i] \right\},$$

where $\mathbf{1}\{\cdot\}$ is an indicator function, and the updated quantities$\{q^0_i\}_{i \in I}$ are based on

$$q^0_i - q^0_{i-1} \in \arg\max \left\{ u_i(q^0_{i-1} + q, T^0(q^0_{i-1}) + \tilde{\tilde{c}}_i q : q \right\},$$

with $q^0_1 = 0$, which are optimal for each type in the equilibrium after the cost increase. Under **Condition IT**, (14), type 1 agent exits the market and the updated aggregate market tariff is

$$T^1(q) \equiv \sum_{i \in I} \tilde{\tilde{c}}_i (q - q^1_{i-1}) \mathbf{1} \left\{ q \in [q^1_{i-1}, q^1_i] \right\},$$

where $q^1_i - q^1_{i-1} \in \arg\max \left\{ u_i(q^1_{i-1} + q, T^1(q^1_{i-1}) + \tilde{\tilde{c}}_i q : q \right\}$ with $q^1_1 = 0$. The exit of agent 1 lowers the utility of each agent because the first layer over the interval $[0, q_1]$ with the lowest tariff, $\tilde{\tilde{c}}_1$, disappears. Also, by the same arguments in the proof of Theorem 1, all the remaining agents are participating in the market with the higher average cost of service. Thus, agents $i > 2$ trading the next available lowest cost interval, $[q_1, q_2]$, suffer further utility declines even without the exit of agent 2. If the new aggregate market tariff, T^1, induces agent 2 to exit, then it must be the case that

$$\max_{q \geq 0} u_2 \left(q, T^1(q) \right) = \max_{q \geq 0} u_2 \left(q, \tilde{\tilde{c}}_2 q \right) \leq \gamma_2,$$

and exiting the market gives higher utility to agent 2.

Now extend the argument recursively to finish the proof by mathematical induction. For an arbitrary $k < i$, suppose that under any candidate equilibrium, agents up to k exit the market. Then, the new market tariff becomes

$$T^k(q) \equiv \sum_{i \in I} \tilde{\tilde{c}}_i(q - q_{i-1}^k)\mathbf{1}\left\{q \in [q_{i-1}^k, q_i^k]\right\},$$

where $q_i^k - q_{i-1}^k \in \arg\max\left\{u_i(q_{i-1}^k + q, T^k(q_{i-1}^k) + \tilde{\tilde{c}}_i q : q\right\}$ with $q_1^k = \cdots = q_k^k = 0$. If agent $k+1$ exits the market under T^k, then

$$\max_{q \geq 0} u_{k+1}\left(q, T^k(q)\right) = \max_{q \geq 0} u_{k+1}\left(q, \tilde{\tilde{c}}_{k+1} q\right) \leq \gamma_{k+1}$$

should hold, because agent $k+1$ will still enter the market otherwise. If the above inequality does not hold, then the equilibrium of active markets is determined starting from $q_{k+1} > 0$ and the initial assumption of $q_{k+1} = 0$ is violated. Therefore, **Condition MS**(i) is necessary.

■

Outside options have two roles. First, they generate discontinuous jumps in market quantities, q_i, after small cost increases. A small cost increase will trigger type i agent to exit with strictly positive quantity q_i when i's utility is close to the utility from the outside option as $u_i(q_i, T(q_i)) \approx \gamma_i$. By contrast, quantities always change smoothly without outside options. Second, outside options generate negative spillovers that can trigger discontinuous exit cascades.[13] The following section more formally presents the exit cascade mechanism.

5.1.1 Description of Cascade of Exits

We provide a formal description of the exit cascade mechanism and general conditions on the size of outside options needed for exit cascades to occur. Assume a cost increase, $\tilde{\tilde{c}}_1$, that is sufficient to ensure condition (14) holds for agent 1 to exit the market. If

$$\max_{q \geq 0} u_2(q, \tilde{\tilde{c}}_2 q) \leq \gamma_2,$$

[13]This mechanism is similar to the worsening adverse selection after a price increase in Stiglitz and Weiss (1981).

agent 2 will also exit. Importantly, agent 2's exit condition above can hold even if agent 2 enters the market conditional on agent 1 remaining:

$$\max_{q\geq 0} u_2(q_1 + q, \tilde{\tilde{c}}_1 q_1 + \tilde{\tilde{c}}_2 q) > \gamma_2,$$

where $q_1 = \arg\max_{q\geq 0} u_1(q, \tilde{\tilde{c}}_1 q)$, because

$$\max_{q\geq 0} u_2(q_1 + q, \tilde{\tilde{c}}_1 q_1 + \tilde{\tilde{c}}_2 q) > \max_{q\geq 0} u_2(q, \tilde{\tilde{c}}_2 q)$$

as long as $q_1 > 0$.

A similar but even more powerful mechanism exists for agent 3's entry/exit decision. The level of agent 3's outside option utility level may be low enough that she will enter the market despite agent 1 exiting while agent 2 enters:

$$\max_{q\geq 0} u_3(q_2 + q, \tilde{\tilde{c}}_2 q_2 + \tilde{\tilde{c}}_3 q) > \gamma_3,$$

where $q_2 = \arg\max_{q\geq 0} u_2(q, \tilde{\tilde{c}}_2 q)$. This is because

$$\max_{q\geq 0} u_3(q_1 + (q_2 - q_1) + q, \tilde{\tilde{c}}_1 q_1 + \tilde{\tilde{c}}_2 (q_2 - q_1) + \tilde{\tilde{c}}_3 q) > \max_{q\geq 0} u_3(q_2 + q, \tilde{\tilde{c}}_2 q_2 + \tilde{\tilde{c}}_3 q) > \max_{q\geq 0} u_3(q, \tilde{\tilde{c}}_3 q),$$

which implies that γ_3 can be between $\max_{q\geq 0} u_3(q_2 + q, \tilde{\tilde{c}}_2 q_2 + \tilde{\tilde{c}}_3 q)$ and $\max_{q\geq 0} u_3(q, \tilde{\tilde{c}}_3 q)$. Extending this argument to any general i, inductively, we see that

$$\begin{aligned}
&\max_{q\geq 0} u_i(q_1 + (q_2 - q_1) + \cdots + (q_{i-1} - q_{i-2}) + q, \tilde{\tilde{c}}_1 q_1 + + \tilde{\tilde{c}}_2 (q_2 - q_1) \cdots + \tilde{\tilde{c}}_{i-1}(q_{i-1} - q_{i-2}) + \tilde{\tilde{c}}_i q) \\
&> \max_{q\geq 0} u_i(q_2 + \cdots + (q_{i-1} - q_{i-2}) + q, \tilde{\tilde{c}}_2 q_2 + \cdots + \tilde{\tilde{c}}_{i-1}(q_{i-1} - q_{i-2}) + \tilde{\tilde{c}}_i q) \\
&\qquad \vdots \\
&> \max_{q\geq 0} u_i(q, \tilde{\tilde{c}}_i q),
\end{aligned}$$

so γ_i can be as low as an element of an ϵ-neighborhood of $\max_{q\geq 0} u_i(q, \tilde{\tilde{c}}_i q)$.

The key insight of the exit cascade argument is that the only outside option value that needs to be close to an agent's maximal attainable utility is that of the best type who exits first. Conditional on the first agent's exit, the downward jumps in utility accumulate, meaning that agent n faces all the utility decreases that arise from agents' $1, 2, \ldots, n-1$ exiting. If we consider a model where the set of eligible participants is endogenous (e.g., agent 1 is the marginal type who is close to being indifferent between entering and exiting the market), the condition that agent 1's outside option utility level be sufficiently close to their ex ante equilibrium utility level arises naturally. Therefore, the required parametric conditions may not be as restrictive as they appear.

5.1.2 Comparative Statics on the Distribution of Types

We now consider how the distribution of types impacts the exit cascade result in Theorem 2. Specifically, how does the coarseness of agent types impact exit cascades? Let $\{I, m, u, c, \gamma\}$ represent an economy where $m = \{m_i\}_{i\in I}$, $u = \{u_i\}_{i\in I}$, $c = \{c_i\}_{i\in I}$, and $\gamma = \{\gamma_i\}_{i\in I}$. Consider another economy, $\{\hat{I}, \hat{m}, \hat{u}, \hat{c}, \hat{\gamma}\}$, that is a *coarser partition* of $\{I, m, u, c, \gamma\}$ if the following holds:

1. $\hat{I} \subset I$.

2. If $i \in \hat{I}$ and $i+1 \in \hat{I}$, then $\hat{m}_i = m_i$, $\hat{u}_i = u_i$, and $\hat{c}_i = c_i$.

3. If $i \in \hat{I}$ and $i+1, \ldots, i+k \notin \hat{I}$, while $i+k+1 \in \hat{I}$ or $i+k+1 > n$, where $k \geq 1$, then agent $i \in \hat{I}$ *includes* agents $i, i+1, \ldots, i+k \in I$ and $\hat{m}_i = \sum_{l=0}^{k} m_{i+l}$, $\hat{u}_i(q,t) = \dfrac{\sum_{l=0}^{k} m_{i+l} u_{i+l}(q,t)}{\sum_{l=0}^{k} m_{i+l}}$, $\hat{c}_i = \dfrac{\sum_{l=0}^{k} m_{i+l} c_{i+l}}{\sum_{l=0}^{k} m_{i+l}}$, and $\hat{\gamma}_i = \dfrac{\sum_{l=0}^{k} m_{i+l}\gamma_{i+l}}{\sum_{l=0}^{k} m_{i+l}}$.

The above definition implies that a coarser partition of an economy groups adjacent types of agents into one type. The mass of the new type of agent is equal to the sum of all masses for each type in the group, and the servicing cost and outside option values are the weighted average cost and outside option values, respectively. The utility of this new agent is the weighted average utility across different types. For example, for $I = \{1, 2, 3, 4, 5\}$ and $\hat{I} = \{1, 2, 4\}$, $\{\hat{I}, \hat{m}, \hat{u}, \hat{c}, \hat{\gamma}\}$ is a coarser partition of $\{I, m, u, c, \gamma\}$, if

$$\begin{aligned}
\hat{m} &= \{m_1, m_2 + m_3, m_4 + m_5\} \\
\hat{u} &= \left\{u_1, \frac{m_2 u_2 + m_3 u_3}{m_2 + m_3}, \frac{m_4 u_4 + m_5 u_5}{m_4 + m_5}\right\} \\
\hat{c} &= \left\{c_1, \frac{m_2 c_2 + m_3 c_3}{m_2 + m_3}, \frac{m_4 c_4 + m_5 c_5}{m_4 + m_5}\right\} \\
\hat{\gamma} &= \left\{\gamma_1, \frac{m_2 \gamma_2 + m_3 \gamma_3}{m_2 + m_3}, \frac{m_4 \gamma_4 + m_5 \gamma_5}{m_4 + m_5}\right\}.
\end{aligned}$$

Note that the upper-tail conditional expected cost is the same in coarse partition as the original partition.

Proposition 3 *Let $\{\hat{I}, \hat{m}, \hat{u}, \hat{c}, \hat{\gamma}\}$ be a coarser partition of $\{I, m, u, c, \gamma\}$ with $i \in \hat{I}$ and $i + 1 \notin \hat{I}$. Then, there exists a cost vector $\tilde{c}$ with $\tilde{c}_j \geq c_j$ for all $j \in I$, such that i does not exit in $\{\hat{I}, \hat{m}, \hat{u}, \hat{\tilde{c}}, \hat{\gamma}\}$, while i exits in $\{I, m, u, \tilde{c}, \gamma\}$. Furthermore, if $i \in \hat{I}$ exits in $\{\hat{I}, \hat{m}, \hat{u}, \hat{\tilde{c}}, \hat{\gamma}\}$, then $i \in I$ exits in $\{I, m, u, \tilde{c}, \gamma\}$.*

Proof. First, we show that if the cost increase to $\tilde{c}$ causes type $i \in \hat{I}$ to exit in the coarser partition $\{\hat{I}, \hat{m}, \hat{u}, \hat{\tilde{c}}, \hat{\gamma}\}$, then agent i exits in the original economy $\{I, m, u, \tilde{c}, \gamma\}$. Suppose that $i \in \hat{I}$, $i + 1, \ldots, i + k \notin \hat{I}$, and $i + k + 1 \in \hat{I}$, i exits the market in the economy $\{\hat{I}, \hat{m}, \hat{u}, \hat{\tilde{c}}, \hat{\gamma}\}$. Thus,

$$\max_{q \geq 0} \left[\frac{m_i u_i(q, \hat{\tilde{c}}_i q) + \cdots + m_{i+k} u_{i+k}(q, \hat{\tilde{c}}_i q)}{m_i + \cdots + m_{i+k}}\right] \leq \hat{\gamma}_i = \frac{m_i \gamma_i + \cdots + m_{i+k} \gamma_{i+k}}{m_i + \cdots + m_{i+k}}$$

holds. Suppose the contrary that i does not exit the market in the economy $\{I, m, u, \tilde{c}, \gamma\}$. Then,

$$\max_{q \geq 0} u_i(q, \hat{\tilde{c}}_i q) = u_i(q_i, \hat{\tilde{c}}_i q_i) > \gamma_i.$$

Thus, for each l, $u_{i+l}(q_i, \hat{\tilde{c}}_i q_i) > \gamma_{i+l}$ holds by (13), where $1 \leq l \leq k$. However, this implies the weighted average of utilities would exceed the weighted average of outside options. In other words,

$$\max_{q\geq 0}\left[\frac{m_i u_i(q,\tilde{\bar{c}}_i q)+\cdots+m_{i+k}u_{i+k}(q,\tilde{\bar{c}}_i q)}{m_i+\cdots+m_{i+k}}\right]\geq \frac{m_i u_i(q_i,\tilde{\bar{c}}_i q_i)+\cdots+m_{i+k}u_{i+k}(q_i,\tilde{\bar{c}}_i q_i)}{m_i+\cdots+m_{i+k}}>\hat{\gamma}_i,$$

which is a contradiction.

Now we show the converse that agent i, who exits the market in the economy $\{I,m,u,\tilde{c},\gamma\}$ due to the cost increase, may not exit in the coarser partition $\{\hat{I},\hat{m},\hat{u},\hat{\tilde{c}},\hat{\gamma}\}$ with the same cost increase. Suppose that $i\in\hat{I}$, $i+1,i+2,\ldots,i+k\notin\hat{I}$, and $i+k+1\in\hat{I}$. Agent i exits the market, if

$$\max_{q\geq 0} u_i(q,\tilde{\bar{c}}_i q)\leq\gamma_i. \tag{16}$$

By continuity of the utility function, maximand (by Berge's maximum theorem), and upper-tail conditional expected cost, the left-hand side of (16) is continuously decreasing in $\tilde{\bar{c}}_i$. Then, there exists $\tilde{c}^*$ such that

$$\max_{q\geq 0} u_i(q,\tilde{\bar{c}}_i^* q)=u_i(q_i,\tilde{\bar{c}}_i^* q_i)=\gamma_i.$$

For each l, $u_{i+l}(q_i,\tilde{\bar{c}}_i^* q_i)>\gamma_{i+l}$ holds by (13), where $l\geq 1$. Again, the utility of $i\in\hat{I}$, the weighted average of utilities, become

$$\begin{aligned}\max_{q\geq 0}\left[\frac{m_i u_i(q,\tilde{\bar{c}}_i^* q)+\cdots+m_{i+k}u_{i+k}(q,\tilde{\bar{c}}_i^* q)}{m_i+\cdots+m_{i+k}}\right]&\geq \frac{m_i u_i(q_i,\tilde{\bar{c}}_i^* q_i)+\cdots+m_{i+k}u_{i+k}(q_i,\tilde{\bar{c}}_i^* q_i)}{m_i+\cdots+m_{i+k}}\\&>\hat{\gamma}_i=\frac{m_i\gamma_i+\cdots+m_{i+k}\gamma_{i+k}}{m_i+\cdots+m_{i+k}},\end{aligned}$$

and type $i\in\hat{I}$ agent does not exit. Hence, there exists a cost vector $\tilde{c}^*$, which makes i to exit in $\{I,m,u,\tilde{c}^*,\gamma\}$ but not in $\{\hat{I},\hat{m},\hat{u},\hat{\tilde{c}}^*,\hat{\gamma}\}$. ■

This result shows that markets are less vulnerable to shutdowns and exit cascades when the partition of types becomes coarser. Conversely, exit cascades are more likely when the type distribution approaches a continuum, since the cost increase needed to trigger market shutdowns falls as the relative mass of each type shrinks. See Appendix C for a visual illustration of the result.

A corollary of Proposition 3 is that markets become more vulnerable to exits as the number of types grows, even with no change in underlying aggregate risk or uncertainty. This highlights why a multi-type model is essential for capturing large swings in trade from small cost increases. A model with only two types, while tractable and useful for illustrating certain forces, does not fully capture the market's vulnerability to exit cascades.

We believe this result is apt to describe the recent CLO-market freezes. CLOs are purchased by an increasing variety of agents, such as banks, insurance companies, private equity funds, mutual funds, etc. Proposition 3 suggests that CLO markets become more susceptible to shutdown with only a small change in fundamentals as the number of distinct participant types increases.

5.2 Implications and Discussions

5.2.1 Policy Implications

The model with outside options generates amplification effects—a small increase in costs can generate a sudden market shutdown. This property resembles real-world financial market movements and has an important policy implication: relatively inexpensive policy interventions can prevent sudden and costly market collapses. For example, if the social planner lowers the market tariff with a total subsidy of $(\tilde{\bar{c}}_1 - \bar{c}_1)q_1 \sum_{i \in I} m_i$, then it is sufficient to prevent type 1 agent's exit and the subsequent exit cascade. This policy is desirable as long as the potential welfare losses, $\sum_{i \in I} (u_i(q_i, T(q_i)) - \gamma_i)$[14], together with the spillovers to other

[14]Since suppliers are competitive, they break even in any equilibrium even under complete breakdown. Therefore, the measure of social welfare is simply the sum of utilities across all types of agents (buyers).

markets, are greater than the intervention cost.[15] Therefore, our model provides a simple yet important reason to support market functioning *even with adverse selection* to prevent more widespread market breakdowns.

The cascade of exits is determined by both the degree of adverse selection and agents' outside options. If entry is very costly—for example, because of high entry or regulatory costs due to heavy usage of balance sheets—the outside option of agents not entering the market could be more profitable. Then, there will be more exits in the market. A moderate reduction of such costs (or reduction of the opportunity cost) could drastically change the allocation by preventing the chain of exits and sudden collapse of the quantities traded in the equilibrium.

5.2.2 Market Monitoring and Systemic Risk

Our results highlight the importance of monitoring the *best (most reliable/least costly) agent* in the market. Even though monitoring the overall condition and the worst (most risky/most costly) participants would be crucial in identifying the increase in the upper-tail conditional expected cost, exits of the most reliable agents are the trigger of the exit cascade. Hence, monitoring the exit incentives of the best participant in each market is important for predicting and preventing shut downs ex ante.

The model does not rely on detailed market structure or other complex agent interactions. It can therefore be applied to various markets and contexts with adverse selection to shed light on how adverse selection can cause partial or full market shutdowns.

Our result that economies with more types (less coarse partitions) are more vulnerable to exit cascades provides a general theoretical underpinning to the information production literature (see for example, Gorton and Ordoñez (2019, 2020); Dang et al. (2020)). In particular, one interpretation is that economies with more types grouped together have more imprecise or opaque information about each individual type. This interpretation is apt

[15] There are many complicated issues related to optimal interventions such as changing incentives under the new rules (or mechanisms) as discussed in Philippon and Skreta (2012).

for models where an agent's type is stochastic as in production economies or asset holdings where agents maximize over the expected value of their types. Our model shows that if more precise information about each agent's type results in more recognizable agents in the economy—a less coarse partition—and more asymmetric information between buyers and sellers, then the market is more unstable. This is in line with Dang et al. (2020) who show that information production can lead to a collapse in the market. Our result shows a similar phenomenon with a more simple, static model.

5.2.3 Microfoundations of Outside Options

Though sufficient, condition (13) is not necessary for exits to generate additional exits due to negative spillovers. However, without some structure, the additional complexity of Rothschild and Stiglitz (1976)-type of non-existence problems arise. In addition, the pattern of exits could become more complicated as there might be no cutoff type, θ, that partitions agents into those who exit and remain. In particular, some intermediate valued agent may remain in the market if the agent's outside option utility is very low.

We introduce two different interpretations of outside options to justify assumption (13). The first interpretation is a fixed entry cost. Suppose that each agent must pay a fixed entry cost given by $\xi > 0$ if they choose to enter the market. Upon entry, the agent trades a positive quantity, q, while paying the market tariff of $T(q)$. The outside option is not paying the entry cost of ξ, or a negative payment with zero trade quantity: $u_i(0, -\xi)$. Then, by continuity and Assumption 2, there exists $\underline{q}_i$ such that

$$u_i(\underline{q}_i, 0) = u_i(0, -\xi) = \gamma_i$$

for any i. Also, by the single-crossing property,

$$u_i(\underline{q}_i, 0) \geq u_i(0, -\xi) \Rightarrow u_j(\underline{q}_i, 0) > u_j(0, -\xi),$$

for any $j > i$. Therefore, $\underline{q}_i$ is decreasing in the index i, and

$$u_i(q,t) \geq \gamma_i \Rightarrow u_j(q,t) > \gamma_j, \qquad \forall i < j,$$

by the single-crossing property. Hence, (13) is a result of the single-crossing condition.

The second way to interpret outside options is to consider the opportunity cost of agents entering a separate market that requires costly verification of agent's type.[16] In this market, agents pay a fixed cost of κ to trade, and the market can verify each agents type. Therefore, agent 1 may be happy to pay κ and get the lowest tariff c_1 for the quantity q_1, whereas agent n would not be happy to pay κ and pay the highest tariff c_n. Thus, whenever agent $j > i$ exits, agent i may also exit.

Finally, one can think of a general equilibrium model in which the utility levels of outside options are endogenously affected by entry/exit decisions of all agents.[17] Our exit cascade results continue to hold even if the value of outside options declines as the best types exit the market. More specifically, exit cascades occur as long as the downward jump in market utility for remaining agents is larger than the decline in value of the outside option. Such general equilibrium interactions are out of the scope of this paper but will be an interesting direction for future research. Along similar lines, changes in outside option values themselves can generate exit cascades even without changes in costs. However, full-fledged comparative statics on the distribution of outside options would be highly complicated due to the strategic complementarity and substitutability issues discussed in Section B.1.

[16]The secondary market structure of agency mortgage-backed securities (MBS) is a good example. A majority of MBS are traded in the to-be-announced (TBA) market, which pools heterogeneous MBS into a few liquid TBA contracts but induces adverse selection. At the same time, traders can trade high-value MBS outside the TBA market in a much less liquid specified-pool (SP) market by specifying the individual CUSIP, but traders pay higher trading cost in the SP market (Huh and Kim, 2021).

[17]The current formulation of outside options under the two interpretations discussed above does not allow the exit decisions of agents to impact the value of others' outside options. For example, the fixed entry fee is independent of other agents' entry and exit decisions. Also, if the outside option "market" can identify the true type of each agent, entry of moldy lemons or other agents into this outside option market will not affect the utility of the lower type agents.

6 Market Shutdowns in the Model of Attar et al. (2014)

In this section we show that the model of Attar et al. (2014) does not generate market shutdowns after a small change in fundamentals, even when extended with generalized outside options.

There is a privately informed seller, corresponding to the demand side of the previous model, who can simultaneously trade with several identical, uninformed buyers, corresponding to the supply side of the previous model. The seller's private information determines her type and can be either L or H with positive probabilities m_L and m_H, respectively. The type i seller's preferences, represented by a utility function u_i, depends only on the bundles (Q, T), where Q is the aggregate quantity she sells to the buyers and T is the aggregate transfer she receives in return.[18] For each i, u_i is continuously differentiable, with $\partial u_i / \partial T > 0$, and u_i is strictly quasiconcave. Hence, type i's marginal rate of substitution of the good for money, which is

$$\tau_i \equiv -\frac{\partial u_i / \partial Q}{\partial u_i / \partial T},$$

is everywhere well defined and strictly increasing along her indifference curves. As in the previous model, $\tau_i(Q, T)$ can be interpreted as type i's marginal cost of supplying a higher quantity, given that she already trades (Q, T). We also assume a strict single-crossing property holds as in the previous model, i.e.,

$$\tau_h(Q, T) > \tau_L(Q, T), \quad \forall (Q, T) \in \mathbb{R}^2.$$

In other words, type H is less eager to sell an additional unit than type L, thus, indifference curves of the two types cross only once.

[18]Following Attar et al. (2014), we refer the informed type as the seller and the uninformed type as the buyers, however, the generality of the model allows the informed type (the seller) to buy the good instead of selling it. In such a case, the quantity becomes negative, i.e., $Q < 0$. This is not the case in any other models we consider in this paper.

There are $n \geq 2$ identical buyers. There are no direct externalities between them, as each buyer cares only about the quantity q she purchases from the seller and the transfer t she makes in return. Each buyer's preferences over individual quantity-transfer bundles (q, t) are represented by a linear profit function $v_i q - t$ if a buyer receives from type i a quantity q and makes a transfer t in return. Buyers value the good from the high type more, i.e., $v_H > v_L$. Therefore, type H provides a more valuable good to the buyers than type L, but at a higher marginal cost. Denote the average quality of the good as $v \equiv m_L v_L + m_H v_H$, where $v_H > v > v_L$.

The seller and buyers play a nonexclusive trading game in which no buyer can control and contract on the trades that the seller makes with other buyers. The extensive form of the game is as follows.

1. Each buyer k proposes a menu of contracts, that is, a set $C^k \subset \mathbb{R}^2$ of quantity-transfer bundles that contains at least the no-trade contract $(0, 0)$.[19]

2. After privately learning her type, the seller selects one contract from each of the menus C^k offered by the buyers.

A pure strategy for type i is a function that maps each menu profile $(C^1, \ldots, C^n)$ into a vector of contracts $((q^1, t^1), \ldots, (q^n, t^n)) \in C^1 \times \cdots \times C^n$. To ensure that type i's optimization problem

$$\max \left\{ u_i \left(\sum_k q^k, \sum_k t^k \right) : (q^k, t_k) \in C^k \text{ for each } k \right\}$$

always has a solution, Attar et al. (2014) assume the buyers' menus C^k are compact sets. Therefore, they use perfect Bayesian equilibrium as the equilibrium concept and focus on pure-strategy equilibria.

For the equilibrium characterization, it is necessary to define first-best quantities, which

[19]This requirement is equivalent to having an outside option of no trade, preventing the seller from being forced to trade with any particular buyer.

Attar et al. (2014) assume the following: for each i, there exists Q_i^* such that $\tau_i(Q_i^*, v_i Q_i^*) = v_i$. In other words, Q_i^* is the efficient quantity for type i to trade at a unit price v_i that gives an aggregate zero profit for the buyers.

Theorems 1 and 2 in Attar et al. (2014) establish that an equilibrium exists if and only if $\tau_L(0,0) \leq v \leq \tau_H(0,0)$. In other words, an equilibrium exists if and only if at a price equal to the average quality of the good v, type L prefers to sell the good, whereas type H prefers to buy it. Moreover, the equilibrium allocations depend on different parameterizations as follows:

a. If $v_L \leq \tau_L(0,0) \leq v \leq \tau_H(0,0) \leq v_H$, all equilibria are pooling, with $Q_L = Q_H = 0$.

b. Otherwise, all equilibria are separating and the following cases occur:

(i) If $\tau_L(0,0) < v_L < v < v_H < \tau_H(0,0)$, then $Q_L = Q_L^* > 0$ and $Q_H = Q_H^* < 0$.

(ii) If $\tau_L(0,0) < v_L < v \leq \tau_H(0,0) \leq v_H$, then $Q_L = Q_L^* > 0$ and $Q_H = 0$.

(iii) If $v_L \leq \tau_L(0,0) \leq v < v_H < \tau_H(0,0)$, then $Q_L = 0$ and $Q_H = Q_H^* < 0$.

The intuition of the equilibrium characterization is as follows. Consider the equilibrium in case a. First, note that there are gains from trade of selling the good for H type if $\tau_H(0,0) \leq v_H$ without asymmetric information, i.e., the seller would prefer to sell the good to the buyers. In contrast, there are gains from trade of buying the good for L type if $v_L \leq \tau_L(0,0)$ without asymmetric information, i.e., the seller would prefer to buy the good from the buyers ($Q_L < 0$). However, due to asymmetric information and nonexclusive contracting, both L and H types would prefer to pretend they are L type to purchase the good at price v_L, while pretending they are H type to sell the good at price v_H, leading to losses for the buyers. However, at a pooling price of v, only L type would prefer to sell at that price, while only H type would prefer to buy at that price. Thus, the only possible equilibrium is the no-trade equilibrium.

Building on this intuition, the intuition for equilibria in case b naturally follows. Consider case b (i). Because L type values the good less than the buyers, there are gains from trade from selling the good up to the efficient quantity $Q_L^* > 0$. In contrast, H type values the good more than the buyers ($v_H < \tau_H(0,0)$), therefore, H type would prefer to buy from the buyers up to the efficient quantity $Q_H^* < 0$. Because $\tau_L(0,0) < v_H$, L type would prefer not to purchase a good at v_H, and because of $v_L < \tau_H(0,0)$, H type would prefer not to sell a good at v_L. This clear separation of preferences allows the buyers' equilibrium contracts to separate the types into either buyers or sellers of the good, so the equilibrium allocations become $(Q_L^*, v_L Q_L^*)$ with $Q_L^* > 0$ and $(Q_H^*, v_H Q_H^*)$ with $Q_H^* < 0$ for L and H type, respectively.

In the following two subsections, we show that neither introducing small cost increases nor adding generalized outside options causes trades to completely unwind to zero. The main reason is that there is no partial pooling in the equilibrium contrary to the equilibrium in Attar et al. (2021). In equilibrium, either both types are already completely pooled into a no-trade contract; or when one type is trading a non-zero quantity, the other type must trade in the opposite side of the market, i.e., one type is a buyer and the other type is a seller (including the possibility of trading zero quantities). Therefore, the terms of trade are not directly linked other than knife-edge cases.

6.1 Cost Increases Comparative Statics

Now we examine the effect of cost increases on the equilibrium quantities in the model of Attar et al. (2014). We first show that small changes in the cost parameters of type L do not affect the equilibrium quantities traded by type H.

Proposition 4 *In the model of Attar et al. (2014), suppose an equilibrium exists. Then, a small change in the cost parameters (an increase in m_L or a decrease in v_L) almost everywhere (except when $\tau_L(0,0) = v$) does not impact the equilibrium trade volume of H type, Q_H.*

Proof. See Appendix A.1. ■

Proposition 4 shows that small changes in the cost of serving type L do not cause the quantity of trade available to type H to fall to zero (i.e., no trade breakdown). Even though Proposition 4 applies to all cases in Theorem 1 and 2 in Attar et al. (2014), including the no-trade equilibrium, the rest of our analysis focuses on cases in which the equilibrium has non-zero trades.

6.2 Generalized Outside Options

We extend the model by adding outside options, which provides utility level of γ_L and γ_H for type L and H, respectively. If a seller chooses the outside option, then the seller cannot trade with any buyer. The extensive form of the game is the following:

1. After privately learning her type, the seller chooses whether to take the outside option or enter the market.
2. Each buyer k proposes a menu of contract C^k, which includes no trade, i.e., $(0, 0)$.
3. If the seller chooses to enter the market, then the seller selects one contract from each of the menus C^k offered by the buyers.

We show that small changes in the cost parameters of type L do not affect the equilibrium trading decision of type H.

Proposition 5 *Suppose informed types have non-trivial outside options $\gamma_i > u_i(0, 0)$, $i = L, H$. Suppose that the preferences of both types are as in one of the separating equilibrium cases (b(i), b(ii), or b(iii)) of Theorem 1 in Attar et al. (2014). Then, a small decrease in v_L does not affect the H type's trading decision, therefore, there are no exit cascades.*

Proof of Proposition 5.

We analyze the effect of a decrease in v_L to $\tilde{v}_L$ on equilibrium in the separating cases of Theorem 1 of Attar et al. (2014). The key observation is that in separating equilibria, the two types face *independent terms of trade* determined by their respective values v_L and v_H.

Case b(i): $\tau_L(0,0) < v_L < v < v_H < \tau_H(0,0)$, with $Q_L = Q_L^* > 0$ (L sells) and $Q_H = Q_H^* < 0$ (H buys).

In equilibrium, buyers offer a contract for L, $(Q_L^*, v_L Q_L^*)$, where $\tau_L(Q_L^*, v_L Q_L^*) = v_L$, and a contract for H, $(Q_H^*, v_H Q_H^*)$, where $\tau_H(Q_H^*, v_H Q_H^*) = v_H$ and $Q_H^* < 0$.

When v_L decreases to $\tilde{v}_L$, the new equilibrium contract for L becomes $(\tilde{Q}_L^*, \tilde{v}_L \tilde{Q}_L^*)$ where $\tau_L(\tilde{Q}_L^*, \tilde{v}_L \tilde{Q}_L^*) = \tilde{v}_L$. The contract for H remains $(Q_H^*, v_H Q_H^*)$ since v_H is unchanged.

First, we check the incentive compatibility (IC) of the two types:

- H's IC: $u_H(Q_H^*, v_H Q_H^*) \geq u_H(\tilde{Q}_L^*, \tilde{v}_L \tilde{Q}_L^*)$

  Since L's bundle became worse and by single-crossing, we have:

  $$u_H(\tilde{Q}_L^*, \tilde{v}_L \tilde{Q}_L^*) < u_H(Q_L^*, v_L Q_L^*).$$

  Therefore H's IC is easier to satisfy. H has *no increased incentive* to deviate.

- L's IC: $u_L(\tilde{Q}_L^*, \tilde{v}_L \tilde{Q}_L^*) \geq u_L(Q_H^*, v_H Q_H^*)$

  For sufficiently small decreases in v_L, this constraint remains satisfied since initially there was slack, $u_L(Q_L^*, v_L Q_L^*) > u_L(Q_H^*, v_H Q_H^*)$, in equilibrium.

Next, we check the effect of outside options:

- L exits if $u_L(\tilde{Q}_L^*, \tilde{v}_L \tilde{Q}_L^*) < \gamma_L$, which is possible.

- H exits if $u_H(Q_H^*, v_H Q_H^*) < \gamma_H$, which is unchanged from before.

Therefore, for sufficiently small decreases in v_L, H's trading decision is completely unaffected. L may exit, but this does not cause H to exit. Thus, there are no cascades.

Case b(ii): $\tau_L(0,0) < v_L < v \leq \tau_H(0,0) \leq v_H$, with $Q_L = Q_L^* > 0$ and $Q_H = 0$.

H is not trading because either (i) $\gamma_H > u_H(Q_H^*, v_H Q_H^*)$ for any positive trading quantity, or (ii) the market structure does not support H's trade.

When v_L decreases to $\tilde{v}_L$ L's contract changes to $(\tilde{Q}_L^*, \tilde{v}_L\tilde{Q}_L^*)$. In contrast, the value v_H and H's potential trading terms are unchanged. Therefore H's decision to not trade remains unchanged if L remains to trade.

If L exits (when $u_L(\tilde{Q}_L^*, \tilde{v}_L\tilde{Q}_L^*) < \gamma_L$), the equilibrium depends on the H's incentive to enter the market and L's incentive to mimic H.

- If $\gamma_L < u_L(Q_H^*, v_H Q_H^*)$, where $Q_H^* > 0$ because of $\tau_H(0,0) < v_H$, then both types exit the market in equilibrium. In other words, because of L type's incentive to mimic H, a contract designed for H would not be traded in equilibrium.

- If $\gamma_L > u_L(Q_H^*, v_H Q_H^*)$ and $\gamma_H < u_H(Q_H^*, v_H Q_H^*)$, then L exits while H enters the market. In other words, if L exits, then a contract designed for H could be traded, as there is no adverse selection problem.

Therefore, a decrease in v_L does not trigger an additional exit of H, i.e., there are no cascades.

Case b(iii): $v_L \leq \tau_L(0,0) \leq v < v_H < \tau_H(0,0)$, with $Q_L = 0$ and $Q_H = Q_H^* < 0$.

L is not trading because $v_L \leq \tau_L(0,0)$, i.e., selling is not profitable.

When v_L decreases to $\tilde{v}_L < v_L$, the condition $\tilde{v}_L < v_L \leq \tau_L(0,0)$ still holds, thus, L remains unwilling to trade (even more so now). H's contract $(Q_H^*, v_H Q_H^*)$ is determined by v_H, which is unchanged, therefore $u_H(Q_H^*, v_H Q_H^*)$ is unchanged, so H's decision is unchanged.

Under any possible setup for outside options, H's incentives and equilibrium contracts are unaffected by the decrease in v_L (see Appendix A.2 for further details). Therefore, there are no cascades.

In all separating equilibria, changes in v_L do not cause H to exit through a cascade mechanism because:

1. H's terms of trade are determined solely by v_H, which is unchanged.

2. In separating equilibria, types face independent pricing.

3. L's exit (if it occurs) does not change H's available contracts or payoffs.

This contrasts sharply with Attar et al. (2021) where partial pooling creates direct linkages: when one type exits a pool, the pooling price changes for all remaining types, creating cascade dynamics. ■

The intuition for Proposition 5 explains why equilibrium structure is central to market fragility. In the separating equilibria of Attar et al. (2014), the two types never trade in the same direction—one type sells (positive quantity) while the other type buys (negative quantity), or one type exits. Thus, each type faces independent terms of trade, i.e., type L trades at unit price v_L, while type H trades at unit price v_H. When v_L decreases, this directly affects only the L type's trading decision and payoff, while the H type continues to face the same price v_H and the same contract terms, so H's decision to trade or exit remains unchanged. Even if the L type exits the market due to the worse terms, L's exit does not change the menu of contracts available to the H type unless it was prevented by L's incentive compatibility condition.

The results in Proposition 5 stand in sharp contrast to the exit cascade results established in Theorem 2 for the Attar et al. (2021) economy. In the extended model of Attar et al. (2021), equilibrium features partial pooling in which multiple types trade on the same side of the market and share common price layers in the convex market tariff. When one type exits, the pooling price adjusts to reflect a new composition of remaining types, worsening the terms of trade for others and triggering further exit cascades. This direct linkage through shared prices is what enables small cost increases to trigger exit cascades. Thus, our results demonstrate that the presence of non-exclusive contracting and generalized outside options is insufficient to generate exit cascades and market shutdowns. Equilibrium requires pooling or cross-subsidization that creates strategic complementarities in exit decisions.

Interestingly, introducing generalized outside options to the model of Attar et al. (2014) reveals a richer and perhaps surprising role beyond simply inducing exit cascades; they can act as a screening mechanism that enables trades previously unavailable.[20] For example, consider the pooling equilibrium case **a** in which neither type trades (both choose $(0, 0)$), because any attempt by buyers to offer contracts leads to adverse selection—both types want to mimic each other's contracts. However, when generalized outside options are introduced, if one type's outside option is sufficiently attractive relative to market trades (including mimicking a contract designed for the other type), it will voluntarily exit and eliminate the adverse selection problem. The remaining type can now trade at their type-specific price, as it is the only type active in the market. Thus, outside options can serve as an endogenous screening device similar to Board and Pycia (2014), as the willingness to forgo market participation reveals the type information, allowing buyers to safely offer contracts designed for the remaining type. Without the shared contract terms and potential cascade of exits shown in Theorem 2 under the extended model of Attar et al. (2021), the introduction of generalized outside options helps to maintain or revive trades, opening markets rather than shutting them down.

7 Conclusion

We characterize the conditions under which small cost increases generate exit cascades and complete market shutdowns in non-exclusive contracting environments. Without outside options, markets are generically robust: as the worst type exits, remaining agents' marginal rates of substitution weakly increase, preserving their incentives to trade. Once outside options are generalized beyond autarky, however, robustness breaks down. We establish that (partial) pooling in equilibrium combined with outside options above the no-trade utility generates a cascade: the exit of one type raises prices across all shared pooling layers, pushing remaining agents below their outside option thresholds and triggering further exits. By contrast, the separating equilibria of Attar et al. (2014) are immune to cascades even

[20] In Appendix B.2, we examine the case of no-trade (pooling) equilibrium in Attar et al. (2014) and show that generalizing outside options can result in an equilibrium with non-zero trades.

with generalized outside options, because types face independent terms of trade and one type's exit cannot worsen another's. We further show that economies with coarser type partitions are less vulnerable: greater pooling of adjacent types requires a larger shock to trigger the initial exit. These results provide a parsimonious mechanism for sudden market shutdowns that requires no institutional details, dynamic structure, or belief-driven runs, and is applicable to insurance, credit, and over-the-counter markets prone to abrupt collapse.

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Online Appendix

A Omitted Proofs

A.1 Proof of Proposition 4

Proof. By Theorem 1 in Attar et al. (2014), equilibrium exists iff $\tau_L(0,0) < v$ when we exclude the knife-edge case of $\tau_L(0,0) = v$.

Suppose that $v_L \leq \tau_L(0,0) < v \leq \tau_H(0,0) \leq v_H$. Then, an increase in m_L or a decrease in v_L lowers v to $\tilde{v}$, which is still less than $\tau_H(0,0)$. Therefore, Q_H remains equal to 0 after the change in cost parameters.

Suppose that $\tau_L(0,0) < v_L < v < v_H < \tau_H(0,0)$. Then, reducing v_L or increasing m_L preserves all the inequalities or it flips only the ordering of $\tau_L(0,0)$ and v_L as $v_L \leq \tau_L(0,0)$. In either case Q_H remains $Q_H^* < 0$.

Suppose that $\tau_L(0,0) < v_L < v \leq \tau_H(0,0) \leq v_H$. Then, reducing v_L or increasing m_L preserves all the inequalities or it flips only the ordering of $\tau_L(0,0)$ and v_L as $v_L \leq \tau_L(0,0)$. In either case Q_H remains zero.

Suppose that $v_L \leq \tau_L(0,0) \leq v < v_H < \tau_H(0,0)$. Then, Q_H remains $Q_H^* < 0$ after the change in cost parameters. ■

A.2 Additional Details for the Proof of Proposition 5

We provide an exhaustive analysis of all the possible details omitted in the proof of Proposition 5.

We first introduce generalized outside options to the cases b. (i) through b. (iii) in Attar et al. (2014) and show the impact on equilibrium trades. In general, there are two trivial extreme cases: (1) both outside options are too low, so they are irrelevant; (2) both outside options are too high, so mechanically induce exits. In the intermediate cases, outside options can act as a screening mechanism. For example, it is possible that H type does not trade in the original equilibrium in Attar et al. (2014), however, the L type exits after an introduction of outside options, eliminating the asymmetric information problem and incentivizing the H type to enter and trade. The following is the exhaustive list of all cases and parameter configurations:

1. If $\tau_L(0,0) < v_L < v < v_H < \tau_H(0,0)$, hence $Q_L = Q_L^* > 0$, $Q_H = Q_H^* < 0$.
   (a) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium trivially remains the same as in that in Theorem 1 of Attar et al. (2014).
   (b) If $\gamma_L > u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with L type exiting and H type trading $Q_H^* < 0$.
   (c) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with H type exiting and L type trading $Q_L^* > 0$.
   (d) If $\gamma_L > u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.

2. If $\tau_L(0,0) < v_L < v \le \tau_H(0,0) \le v_H$, hence $Q_L = Q_L^* > 0$, $Q_H = 0$.

   (a) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, where $Q_H^* > 0$ such that $\tau_H(Q_H^*, Q_H^* v_H) = v_H$. Then, the equilibrium trivially remains the same as in that in Theorem 1 of Attar et al. (2014).

   (b) If $u_L(Q_L^*, Q_L^* v_L) < \gamma_L < u_L(Q_H^*, Q_H^* v_H)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.

   (c) If $u_L(Q_L^*, Q_L^* v_L) < u_L(Q_H^*, Q_H^* v_H) < \gamma_L$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is a separating equilibrium with L type exiting and H type trading $Q_H^* > 0$.

   (d) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with H type exiting and L type trading $Q_L^* > 0$.

   (e) If $\gamma_L > u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.

3. If $v_L \le \tau_L(0,0) \le v < v_H < \tau_H(0,0)$, hence $Q_L = 0$ and $Q_H = Q_H^* < 0$.

   (a) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$, where $Q_L^* < 0$ such that $\tau_L(Q_L^*, Q_L^* v_L)$, and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, where $Q_H^* < 0$ such that $\tau_H(Q_H^*, Q_H^* v_H) = v_H$. Then, the equilibrium trivially remains the same as in that in Theorem 1 of Attar et al. (2014).

   (b) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$ and $u_H(Q_H^*, Q_H^* v_H) < \gamma_H < u_H(Q_L^*, Q_L^* v_L)$, then the equilibrium is the pooling equilibrium with both types exiting the market.

   (c) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$ and $u_H(Q_H^*, Q_H^* v_H) < u_H(Q_L^*, Q_L^* v_L) < \gamma_H$, then the equilibrium is a separating equilibrium with L type trading $Q_L^* < 0$ and H type exiting the market.

   (d) If $\gamma_L > u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with L type exiting and H type trading $Q_H^* < 0$.

   (e) If $\gamma_L > u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.

Now, we change v_L to $\tilde{v}_L$ and examine what happens to the equilibrium in every case listed above.

1. If $\tau_L(0,0) < v_L < v < v_H < \tau_H(0,0)$, hence $Q_L = Q_L^* > 0$, $Q_H = Q_H^* < 0$. In any of the cases, a decrease in v_L to $\tilde{v}_L$ may induce an exit of L type, but the H's decision remains the same.

   (a) If $\gamma_L < u_L(Q_L^*, Q_L^* \tilde{v}_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium trivially remains the same as in that in Theorem 1 of Attar et al. (2014).

   (b) If $\gamma_L > u_L(Q_L^*, Q_L^* \tilde{v}_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with L type exiting and H type trading $Q_H^* < 0$.

   (c) If $\gamma_L < u_L(Q_L^*, Q_L^* \tilde{v}_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with H type exiting and L type trading $Q_L^* > 0$.

(d) If $\gamma_L > u_L(Q_L^*, Q_L^* \tilde{v}_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.

2. If $\tau_L(0,0) < v_L < v \leq \tau_H(0,0) \leq v_H$, hence $Q_L = Q_L^* > 0$, $Q_H = 0$. In any of the cases, a decrease in v_L to $\tilde{v}_L$ may induce an exit of L type, but the H's decision remains the same.

   (a) If $\gamma_L < u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, where $Q_H^* > 0$ such that $\tau_H(Q_H^*, Q_H^* v_H) = v_H$. Then, the equilibrium trivially remains the same as in that in Theorem 1 of Attar et al. (2014).

   (b) If $u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L) < \gamma_L < u_L(Q_H^*, Q_H^* v_H)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.

   (c) If $u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L) < u_L(Q_H^*, Q_H^* v_H) < \gamma_L$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is a separating equilibrium with L type exiting and H type trading $Q_H^* > 0$.

   (d) If $\gamma_L < u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with H type exiting and L type trading $Q_L^* > 0$.

   (e) If $\gamma_L > u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.

3. If $v_L \leq \tau_L(0,0) \leq v < v_H < \tau_H(0,0)$, hence $Q_L = 0$ and $Q_H = Q_H^* < 0$. Again, in any of the cases, a decrease in v_L to $\tilde{v}_L$ does not induce an exit of H.

   (a) If $\gamma_L < u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$, where $\tilde{Q}_L^* < 0$ such that $\tau_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$, and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, where $Q_H^* < 0$ such that $\tau_H(Q_H^*, Q_H^* v_H) = v_H$. Then, the equilibrium trivially remains the same as in that in Theorem 1 of Attar et al. (2014).

   (b) If $\gamma_L < u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $u_H(Q_H^*, Q_H^* v_H) < \gamma_H < u_H(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$, then the equilibrium is the pooling equilibrium with both types exiting the market.

   (c) If $\gamma_L < u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $u_H(Q_H^*, Q_H^* v_H) < u_H(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L) < \gamma_H$, then the equilibrium is a separating equilibrium with L type trading $\tilde{Q}_L^* < 0$ and H type exiting the market.

   (d) If $\gamma_L > u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with L type exiting and H type trading $Q_H^* < 0$.

   (e) If $\gamma_L > u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.

B Additional Results

B.1 Rothschild-Stiglitz Unraveling: Nonexistence

This subsection revisits the existence arguments of Attar et al. (2021) in light of introducing outside options. Recall in their model, equilibrium always exists and is unique. We now show that our generalization of outside options also alters the existence result of the model, similar to the Rothschild and Stiglitz (1976) unraveling argument with exclusive contracting.

A necessary condition for the non-existence of equilibrium is that assumption (13) does not hold.[21] The following simple example highlights how outside options can cause equilibrium to break down when (13) does not hold. Suppose that type n agent exits the market because the aggregate market tariff T is too high compared to n's outside option, i.e. $\max_q u_n(q, T(q)) \leq \gamma_n$. However, the exit of n decreases T to $T^\dagger$, which may make n want to enter again. Then, the entry of n will change (increase) $T^\dagger$ to T again, which will make n want to exit again.

Furthermore, an agent's entry or exit could affect other agents' entry and exit decisions, further complicating equilibrium existence. Suppose the cost vector increases to $\tilde{c} > c$. Without loss of generality, we assume that all agents enter the market in the equilibrium before the cost increase. As in the previous section, the new upper-tail conditional expected cost is given by (11). Denote equilibrium quantities by $\tilde{q}_i$ for each i that enters the market where

$$\tilde{q}_i - \tilde{q}_{i-1} \in \arg\max_q \left\{ u_i \left(\tilde{q}_{i-1} + q, \tilde{T}(\tilde{q}_{i-1}) + \tilde{\bar{c}}_i q \right) : q \right\},$$

and $\tilde{T}(q)$ is the equilibrium aggregate affine tariff with slope $\tilde{\bar{c}}_i$ over $[\tilde{q}_{i-1}, \tilde{q}_i]$. In addition, set $\tilde{q}_0 = \tilde{q}_j = 0$ for any j who exits the market.

In this case, agent entry and exit decisions in equilibrium may be either strategic complements or substitutes à la Bulow et al. (1985). Suppose that type i agent does not enter the market in equilibrium, and for simplicity, suppose that all other agents do enter the market. The rest of the arguments go through even if there are other agents who exit the market. Since i exits the market, $\tilde{q}_i = 0$ and

$$\tilde{T}(\tilde{q}_{i+1}) - \tilde{T}(\tilde{q}_{i-1}) = \tilde{\bar{c}}_{i+1} \left(\tilde{q}_{i+1} - \tilde{q}_{i-1} \right).$$

Now compare that to a hypothetical equilibrium with the same quantities $(\tilde{q}_j)_{j<i}$ and corresponding market tariff $\tilde{T}'$ in which type i enters the market. Hence, $\tilde{q}_i > \tilde{q}_{i-1}$ and

$$\tilde{T}'(\tilde{q}_{i+1}) - \tilde{T}'(\tilde{q}_{i-1}) = \tilde{\bar{c}}_{i+1} \left(\tilde{q}_{i+1} - \tilde{q}_i \right) + \tilde{\bar{c}}_i \left(\tilde{q}_i - \tilde{q}_{i-1} \right) < \tilde{\bar{c}}_{i+1} \left(\tilde{q}_{i+1} - \tilde{q}_{i-1} \right),$$

implying that $i+1$ will have higher utility with these quantities and market tariff. The same logic that utilities are increasing when lower types are present applies to all agents greater than i. Because their utilities in the market increase, agents with type greater than i are more likely to enter the market when i enters the market. Therefore, an entry of a lower type agent generates one-way strategic complementarity to higher type agents.

[21] Theorem 1 of Attar et al. (2014) shows that an equilibrium exists only if the adverse selection problem is severe enough, implying that different types' incentives are not too closely aligned.

However, the effect can go the other way for agent j with type lower than i, i.e. $j < i$. If the upper-tail conditional expected cost of agent j excluding i is less than $\tilde{c}_i$—that is,

$$\frac{\tilde{T}(\tilde{q}_j) - \tilde{T}(\tilde{q}_{j-1})}{\tilde{q}_j - \tilde{q}_{j-1}} = \frac{\sum_{k \geq j, k \neq i} m_k \tilde{c}_k}{\sum_{k \geq j, k \neq i} m_k} < \tilde{c}_i,$$

then the entry of i will increase the upper-tail conditional expected cost for agents with type j and lower, as the new upper-tail conditional expected cost is

$$\frac{\sum_{k \geq j, k \neq i} m_k}{\sum_{k \geq j} m_k} \frac{\sum_{k \geq j, k \neq i} m_k \tilde{c}_k}{\sum_{k \geq j, k \neq i} m_k} + \frac{m_i}{\sum_{k \geq j} m_k} \tilde{c}_i,$$

which is a convex combination of $\dfrac{\tilde{T}(\tilde{q}_j) - \tilde{T}(\tilde{q}_{j-1})}{\tilde{q}_j - \tilde{q}_{j-1}}$ and $\tilde{c}_i$, so

$$\frac{\tilde{T}(\tilde{q}_j) - \tilde{T}(\tilde{q}_{j-1})}{\tilde{q}_j - \tilde{q}_{j-1}} = \frac{\sum_{k \geq j, k \neq i} m_k \tilde{c}_k}{\sum_{k \geq j, k \neq i} m_k} < \frac{\sum_{k \geq j, k \neq i} m_k}{\sum_{k \geq j} m_k} \frac{\sum_{k \geq j, k \neq i} m_k \tilde{c}_k}{\sum_{k \geq j, k \neq i} m_k} + \frac{m_i}{\sum_{k \geq j} m_k} \tilde{c}_i.$$

Because of this cost increase, agent j's maximum utility in the market decreases, and j may exit the market. Hence, entry of higher types can generate one-way strategic substitutability to a lower type. Therefore, the overall effect of the equilibrium remains ambiguous. Thus, the outside options generate key interactions between entry/exit decisions of different agents that can destabilize the otherwise very robust and stable entry-proof equilibrium of Attar et al. (2021). Levy and Veiga (2021) show a similar result in a competitive exclusive contracting environment when the cost c_n is unbounded (or tends to infinity). By contrast, with non-exclusive contracting, our non-existence of equilibrium does not require infinite or unbounded costs.

However, depending on distributional assumptions, outside options also generate the cascade of exits from a small change in fundamentals as in subsection 5.1. Specifically, condition (13) generates Proposition 2, which is important because an exit of the lowest (best) type agent always has an unambiguous, negative, and strategic effect to higher type agents—exits of the lower type agents generate additional exits. Therefore, an additional assumption that provides structure to the distribution of the outside option values is needed to guarantee that the equilibrium exists.

B.2 Generalized Outside Options in No-Trade Equilibrium of Attar et al. (2014)

This subsection examines the role of generalized outside options in the no-trade (pooling) equilibrium of Attar et al. (2014)—that is, both types trade the no-trade contract $(0, 0)$. Proposition B.1 shows that we can revive trade when we generalize outside options, but Proposition B.2 shows that such a revived equilibrium may go back to no-trade equilibrium after a small decrease in v_L.

Proposition B.1 *Suppose informed types have non-trivial outside options equal to $\gamma_i > u_i(0,0), i = L, H$. Suppose that the preferences of both types are as in the case of pool-*

ing equilibrium in Theorem 1 of Attar et al. (2014). Then, the equilibrium is separating equilibrium if either

1. *$\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$ and $\gamma_H \geq u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$, where $\tilde{Q}_L < 0$ such that $\tau_L(\tilde{Q}_L, \tilde{Q}_L v_L) = v_L$; or*

2. *$\gamma_H < u_H(\tilde{Q}_H, \tilde{Q}_H v_H)$ and $\gamma_L \geq u_L(\tilde{Q}_H, \tilde{Q}_H v_H)$, where $\tilde{Q}_H > 0$ such that $\tau_H(\tilde{Q}_H, \tilde{Q}_H v_H) = v_H$.*

In all other cases, the equilibrium remains to be pooling equilibrium in which both types choose the outside option.

Proof of Proposition B.1.
Step 1.
Suppose that both types L and H enter the market. Then, the only available contract is the no-trade contract with $(0, 0)$. Hence, both types prefer the outside option over staying in the market. Therefore, there is no equilibrium in which both types enter the market.

Step 2.
Case 1. Suppose that $\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$, where $\tilde{Q}_L < 0$ is such that $\tau_L(\tilde{Q}_L, \tilde{Q}_L v_L) = v_L$. Suppose that H takes the outside option and L remains in the market. Then, $v_L \leq \tau_L(0, 0)$ implies that L type would buy at the price v_L, i.e., $\tilde{Q}_L < 0$.

Case 1.1. If $\gamma_H \geq u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$, then equilibrium is separating with $\tilde{Q}_L < 0, \tilde{Q}_H = 0$, because H type would prefer to take the outside option over the market trade.

Case 1.2. If $\gamma_H < u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$, then H type also prefers to trade the contract $(\tilde{Q}_L, \tilde{Q}_L v_L)$ in the market and would like to enter. However, if both types enter the market, then Theorem 1 in Attar et al. (2014) implies that the only possible equilibrium contract is $(0, 0)$, which cannot be an equilibrium as we discussed in Step 1. Therefore, there is no equilibrium with H type exits and L type enters, if $\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$ and $\gamma_H < u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$.
Case 2. Suppose that $\gamma_H < u_H(\tilde{Q}_H, \tilde{Q}_H v_H)$, where $\tilde{Q}_H > 0$ is such that $\tau_H(\tilde{Q}_H, \tilde{Q}_H v_H) = v_H$. Suppose that L type exits and H remains in the market. Then, $\tau_H(0, 0) \leq v_H$ implies that H type would sell at the price v_H, i.e., $\tilde{Q}_H > 0$.

Case 2.1. If $\gamma_L \geq u_L(\tilde{Q}_H, \tilde{Q}_H v_H)$, then equilibrium is separating with $Q_L = 0, \tilde{Q}_H > 0$, because L type would prefer to take the outside option over the market trade.

Case 2.2. If $\gamma_L < u_L(\tilde{Q}_H, \tilde{Q}_H v_H)$, then L type also prefers to trade the contract $(\tilde{Q}_H, \tilde{Q}_H v_H)$ in the market and would like to enter. However, if both types enter the market, then Theorem 1 in Attar et al. (2014) implies that the only possible equilibrium contract is $(0, 0)$, which cannot be an equilibrium as we discussed in Case 1. Therefore, there is no equilibrium with H type enters and L type exits, if $\gamma_H < u_H(\tilde{Q}_H, \tilde{Q}_H v_H)$ and $\gamma_L < u_L(\tilde{Q}_H, \tilde{Q}_H v_H)$.
Case 3. If $\gamma_H \geq u_H(\tilde{Q}_H, \tilde{Q}_H v_H)$ and $\gamma_L \geq u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$, then both L and H type exit the market. ■

Proposition B.1 implies that we can revive trades in the market even in the case in which there were no trades in Attar et al. (2014). Outside options can be effective in screening out one type from mimicking the other type, as the utility from the outside options can exceed the payoff from mimicking the market trade of the other type. However, this result depends on the values of the outside options.

One last thing to note is that there could be self-fulfilling no trade equilibrium. Even if $\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$ and $\gamma_H < u_h(\tilde{Q}_H, \tilde{Q}_H v_H)$, if the uninformed investors believe that both types are exiting, they are indifferent between offering any contract, including the no-trade contract $(0, 0)$. If the no-trade contract is the only contract being offered, both types would prefer to exit and take their outside option. However, we can rule out this self-fulfilling no-trade equilibrium by simple equilibrium refinements such as the trembling-hand equilibrium or sequential equilibrium. For the simplicity of exposition, we do not go into this equilibrium refinement and assume that there is no self-fulfilling no-trade equilibrium.

The next proposition shows that such an equilibrium with positive (negative) trades can break down after a small changes in the values (costs) for uninformed buyers, when the utility levels of the outside options are close to the edge of mimicking the other type. Under such a case, the equilibrium goes back to the no-trade equilibrium.

Proposition B.2 *Suppose that the conditions in Proposition B.1 hold. A small decrease in v_L can make an equilibrium with positive trades to breakdown to a no-trade equilibrium.*

Proof.
Case 1. Suppose that $\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$, where $\tilde{Q}_L < 0$ is such that $\tau_L(\tilde{Q}_L, \tilde{Q}_L v_L) = v_L$. Now consider the case of v_L decreasing to $\hat{v}_L$.

Case 1.1. Suppose that the equilibrium is a separating equilibrium because L type enters, while H type exits, i.e., $\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$ and $\gamma_H \geq u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$. If v_L decreases to $\hat{v}_L$, then L type still prefers to trade in the market with $\hat{\tilde{Q}}_L < \tilde{Q}_L$ such that $\tau_L(\hat{\tilde{Q}}_L, \hat{\tilde{Q}}_L \hat{v}_L) = \hat{v}_L$. However, the inequality for the H type can be reversed as $u_H(\hat{\tilde{Q}}_L, \hat{\tilde{Q}}_L \hat{v}_L) > \gamma_H \geq u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$. Thus, H type now has an incentive to mimic the L type, implying that the uninformed buyers will only suggest no trade contracts, leading to both types choosing the outside option over the market trade.

Case 1.2. Suppose that the equilibrium is a pooling equilibrium with both types choosing the outside option because $\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$ and $\gamma_H < u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$, i.e., the H type would like to mimic the L type if there is a contract designed for the L type (Case 1.2. in the proof of Proposition B.1). Then, the unit price of the contract $\hat{v}_L$ is lower than before, so both inequalities remain the same as

$$\gamma_L < u_L(\hat{\tilde{Q}}_L, \hat{\tilde{Q}}_L \hat{v}_L)$$
$$\gamma_H < u_H(\hat{\tilde{Q}}_L, \hat{\tilde{Q}}_L \hat{v}_L),$$

therefore, the equilibrium is still a pooling equilibrium with both types choosing the outside option.
Case 2. Suppose that $\gamma_H < u_H(\tilde{Q}_H, \tilde{Q}_L v_H)$, where $\tilde{Q}_H > 0$ is such that $\tau_H(\tilde{Q}_H, \tilde{Q}_H v_H) = v_H$. Now consider the case of v_L decreasing to $\hat{v}_L$.

Case 2.1. Now with the decreased valuation from the uninformed buyers, the payoffs from market trade for both types decrease. Therefore, L type still prefers the outside option over the market trade. Hence, the equilibrium remains a separating equilibrium, unless H type was on a knife-edge case of $\gamma_H = u_H(\tilde{Q}_H, \tilde{Q}_L v_H)$.

Case 2.2. Now with the decreased payoff from the market trade, then the L type's outside option can dominate the payoff from mimicking the H type in the market (which is again a knife-edge case). Then, after the decrease in v_H, the equilibrium can switch from a pooling equilibrium to a separating equilibrium with H type buying the good in the quantity of $\hat{\tilde{Q}}_H > 0$, and L type choosing the outside option.
Case 3. In a knife-edge case of Case 3 in the proof of Proposition B.1, $\gamma_L = u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$, a decrease in v_L can make the equilibrium a separating equilibrium if $\gamma_H \geq u_H(\hat{\tilde{Q}}_L, \hat{\tilde{Q}}_L \hat{v}_L)$, i.e. L type purchases the good in the market, and H type chooses the outside option.

A decrease in v_H doesn't affect the pooling equilibrium, as it decreases the payoffs from potential market trades, further disincentivizing both types from entering the market. ∎

C Numerical Example

This section provides a numerical example to highlight the main analytical results. In particular, consider the canonical insurance market with binary loss model of Rothschild and Stiglitz (1976) and Hendren (2014) following the idea of Dubey and Geanakoplos (2019).[22] We first compare the results between the baseline model of Attar et al. (2021) with and without outside options. Then, we consider the comparative statics on the partition of types.

C.1 Model Setup and Equilibrium

Agents have initial endowment $(e, 0)$ for (e_g, e_b), where e_g and e_b represent good state and bad state endowment, respectively. They will consume $(x_g, x_b) = (e, 0)$ under autarky, where x_g and x_b represent good state and bad state consumption, respectively. Suppose that agents have constant relative risk aversion (CRRA) and agent i's utility function is

$$v_i(x_g, x_b) = p_i \log(1 + x_b) + (1 - p_i) \log(1 + x_g),$$

where p_i is the probability of agent i faces a loss and receives bad state endowment. The marginal rate of substitution for agent i is

$$\tau_i(x_g, x_b) \equiv \frac{\dfrac{\partial v_i(x_g, x_b)}{\partial x_b}}{\dfrac{\partial v_i(x_g, x_b)}{\partial x_g}} = \frac{p_i}{1 - p_i} \frac{1 + x_g}{1 + x_b}.$$

Following the assumption of competitive suppliers in the main model, the service cost for suppliers is $c_i = p_i$, so they need p_i amount of x_g to insure $1 - p_i$ amount of x_b to agent i. Under non-exclusive contracts, suppliers require the upper-tail conditional expected cost,

$$\bar{c}_i = \sum_{j \geq i} \frac{m_j p_j}{\sum_{j \geq i} m_j},$$

where m_i is the relative mass of agent i.

Denote the additional utility level of taking the outside option on top of utility under no trade, $v_i(e, 0)$, as γ for each i. Agents will compare the level of utility they can get from their optimal consumption bundle in the market to the level of utility they can get from the outside option and decide whether to enter or exit the market. Utilizing the results on the general model, the **Condition MS(i)** for this model is

$$v_i(x_g^i, x_b^i) \leq \gamma_i = v_i(e, 0) + \gamma,$$

where $v_i(e, 0) + \gamma$ is decreasing in i. If the above inequality holds, then any agent $j \leq i$ exits the market and we set $x_b^j = 0$.

[22] Note that the baseline model of Attar et al. (2021) and our extension with outside options in this paper are much more general and can be applied to many other contexts.

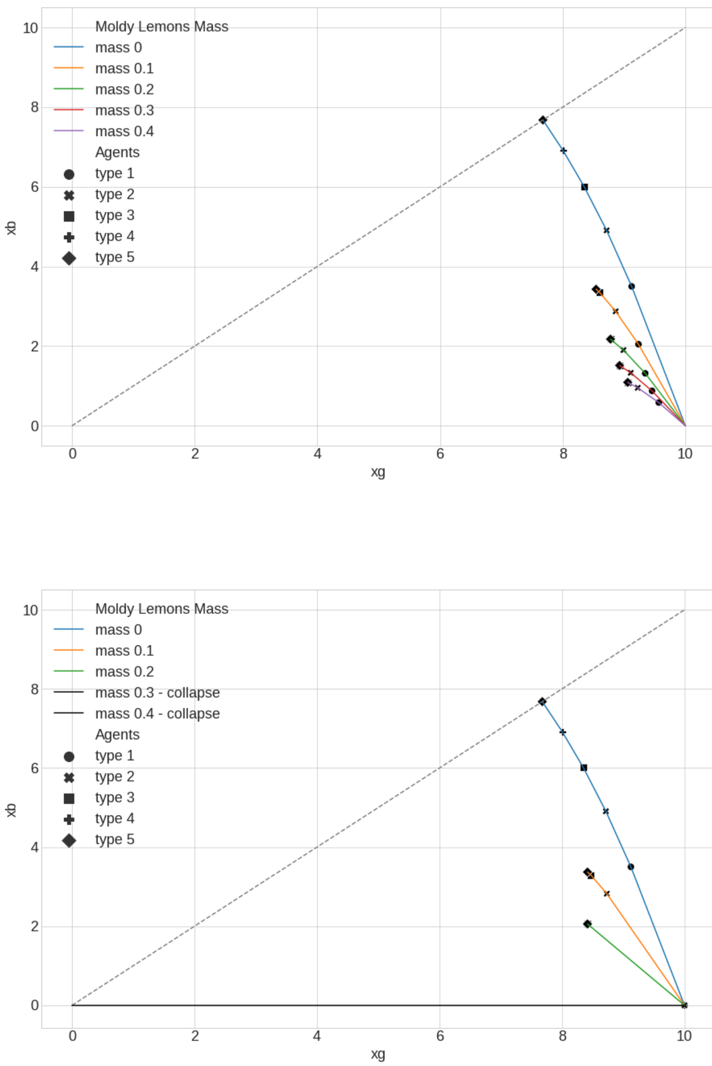

Figure C.1: Consumption bundles of models without and with outside options

Note: Each curve represents consumption possibility frontier and consumption bundles of each agent represented by different shape of dot for a different mass of moldy lemons (the new worst type agent).

C.2 Effect of Outside Options and Cost Increases

We verify the main result of this paper with the numerical model. The cost increase we consider arises from entry of a worst type of agent, which we call a "moldy lemon". This cost increase is a natural case to consider and highlights the amplification result. Denote the cost of supplying moldy lemons as c_{n+1} and the mass of moldy lemons as m_{n+1}. Figure C.1 shows equilibrium of the baseline model in the top panel and equilibrium of the model with outside options in the bottom panel. For each panel, the horizontal axis represents consumption in the good state, x_g, and the vertical axis represents consumption in the bad state, x_b.

Each colored curve represents the consumption possibility frontier for a given mass of moldy lemons in the market. Different shapes of dots represent each agent's optimal consumption bundle given mass of moldy lemons and other agents' consumption in the equilibrium.

First, the results show that x_b decreases monotonically with the increase in the mass of moldy lemons. As more moldy lemons enter the market, the upper-tail conditional expected cost $\tilde{\tilde{c}}_i$ increases, depressing the consumption possibility frontier for every agent in the market. With the higher cost, agents purchase less (insurance) contracts.[23]

Second, the results show a market shutdown in the model with outside options, which does not exist in the model without outside options. Even though total quantities decrease in both models, the model without outside options shows a smooth contraction of quantities traded as the mass of moldy lemons rises. Thus, even when the moldy lemons mass is 0.4, all agents still enter the market and trade in positive amounts. In contrast, the model with outside option shows the exit of agents and total market shutdown at the moldy lemons mass of 0.3. Thus, the numerical exercise shows how the existence of outside options can generate complete market shutdowns even for a small change in costs due to a small entry mass of moldy lemons.[24]

C.3 Coarse Partition of Types and Cost Increases

Using the model with outside options, we analyze comparative statics of Proposition 3 on the division of types depicted in Figure C.2. In particular, we group adjacent types of agents into a single type whose service cost is the weighted average of individual type's cost, and a mass equal to the sum of each type's mass. In particular, we combine type 1 and 2 together to create a new type 1 agent with probability and mass as

$$\hat{p}_1 = \frac{m_1 p_1 + m_2 p_2}{m_1 + m_2}$$
$$\hat{m}_1 = m_1 + m_2.$$

Similarly, we also combine type 3 and 4 in the baseline case into a new type 2 agent as

$$\hat{p}_2 = \frac{m_3 p_3 + m_4 p_4}{m_3 + m_4}$$
$$\hat{m}_2 = m_3 + m_4,$$

while simply renaming the previous agent 5 as agent 3. Therefore, the market-wide uncertainty and service cost remain the same as before.

The top panel of Figure C.2 is the baseline case as shown in the bottom panel of Figure C.1, while the bottom panel of Figure C.2 is the case with a more coarse partition. The numerical results show that the economy with a coarser partition of types is less vulnerable due to cost increases arising from the entry of moldy lemons–there are fewer exits across

[23] This decrease in quantities may not always be the case in the general model, because the income effects might dominate the substitution effects. The functional form of this exercise precludes the case of insurance being a Giffen good.

[24] The definition of complete market shutdown here is the exit of all agents except for moldy lemons.

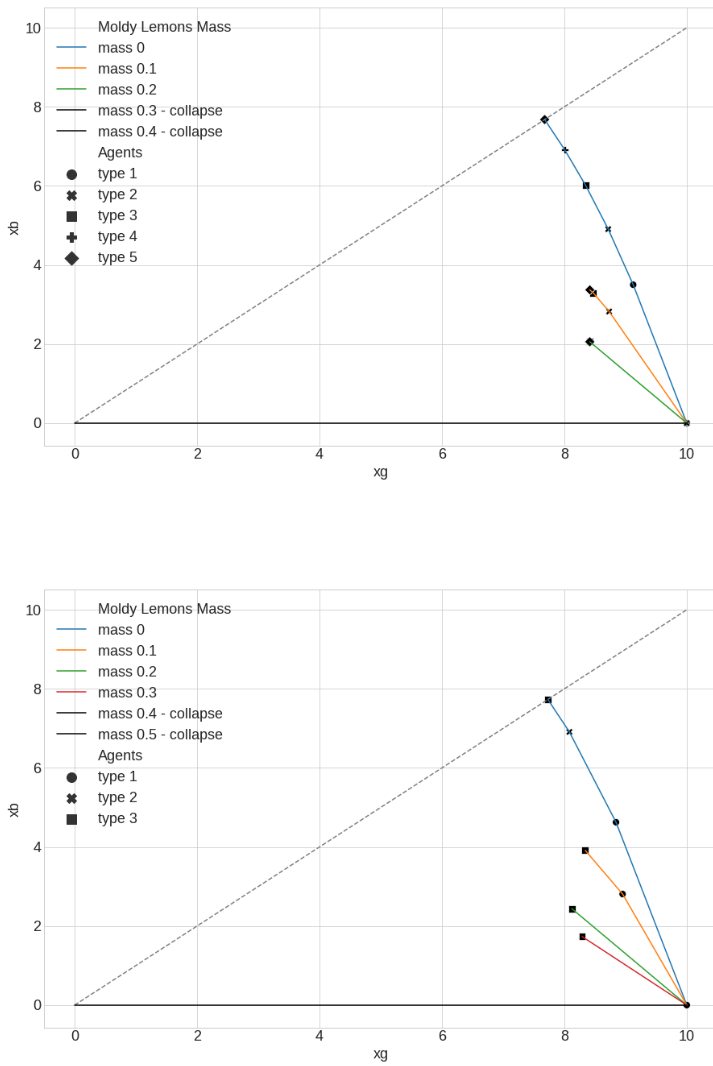

Figure C.2: Consumption bundles of models with different partition of types

Note: Each curve represents consumption possibility frontier and consumption bundles of each agent represented by different shape of dot for a different mass of moldy lemons.

varying masses of moldy lemons. Also, full market shutdown does not happen even when the moldy lemons mass is 0.3, which generated a full market collapse under the baseline case.

C.4 Details of Numerical Simulations

For computational tractability, we would like to represent agents' optimization problem as isomorphic optimization problem across all agents by adjusting the endowments e^i for each agent i. This can be done by exploiting the single-crossing property and any agent $i > j$ would consume as much as agent j does in equilibrium. For a given e^i, the optimal consumption bundle is

$$x_g^* = (1-p_i)e^i - \frac{p_i - \bar{c}_i}{1-\bar{c}_i}$$
$$x_b^* = \frac{p_i - \bar{c}_i}{\bar{c}_i} + \frac{(1-\bar{c}_i)p_i}{\bar{c}_i}e^i,$$

for an interior solution. If it is a corner solution, then $(x_g^*, x_b^*) = \left(0, \frac{1-\bar{c}_i}{\bar{c}_i}e^i\right)$ or $(x_g^*, x_b^*) = (e^i, 0)$.

From the results in the general model, we know that agent 1 first decides on the optimal quantity q_1^* for the given price $\bar{c}_1$ and then agent 2 decides on q_2^* for the price $\bar{c}_2$ on top of purchasing q_1^*, and so forth. Agent 1's optimal consumption bundle is

$$x_g^1 = (1-p_1)e - \frac{p_1 - \bar{c}_1}{1-\bar{c}_1}$$
$$x_b^1 = \frac{p_1 - \bar{c}_1}{\bar{c}_1} + \frac{(1-\bar{c}_1)p_1}{\bar{c}_1}e,$$

assuming that they have interior solutions without loss of generality. For agent 2, the same optimality condition should hold, but the budget constraint is different from the previous representation. This is because agent 2 can purchase the bundle in a cheaper price $\frac{\bar{c}_1}{1-\bar{c}_1}$ instead of $\frac{\bar{c}_2}{1-\bar{c}_2}$ up to x_b^1. Therefore, the updated budget constraint for agent 2 becomes

$$\begin{aligned} x_g^2 &= e - \frac{\bar{c}_1}{1-\bar{c}_1}x_b^1 - \frac{\bar{c}_2}{1-\bar{c}_2}(x_b^2 - x_b^1) \\ &= e + \left(\frac{\bar{c}_2}{1-\bar{c}_2} - \frac{\bar{c}_1}{1-\bar{c}_1}\right)x_b^1 - \frac{\bar{c}_2}{1-\bar{c}_2}x_b^2. \end{aligned}$$

Therefore, we simply change the endowment from e to

$$e^2 = e + \left(\frac{\bar{c}_2}{1-\bar{c}_2} - \frac{\bar{c}_1}{1-\bar{c}_1}\right)x_b^1$$

for the agent 2's budget constraint. Thus, the optimal consumption bundle for agent 2 is

$$x_g^2 = (1-p_2)e^2 - \frac{p_2 - \bar{c}_2}{1-\bar{c}_2}$$
$$x_b^2 = \frac{p_2 - \bar{c}_2}{\bar{c}_2} + \frac{(1-\bar{c}_2)p_2}{\bar{c}_2}e^2.$$

Agent 3's problem is isomorphic to agent 2's problem except that agent 3's endowment is

$$e^3 = e + \left(\frac{\bar{c}_2}{1-\bar{c}_2} - \frac{\bar{c}_1}{1-\bar{c}_1}\right)x_b^1 + \left(\frac{\bar{c}_3}{1-\bar{c}_3} - \frac{\bar{c}_2}{1-\bar{c}_2}\right)x_b^2,$$

and agent 3's optimal consumption bundle becomes

$$x_g^3 = (1 - p_3)e^3 - \frac{p_3 - \bar{c}_3}{1 - \bar{c}_3}$$
$$x_b^3 = \frac{p_3 - \bar{c}_3}{\bar{c}_3} + \frac{(1 - \bar{c}_3)p_3}{\bar{c}_3} e^3.$$

For a general agent i, agent i's updated endowment is

$$e^i = e + \sum_{j<i} \left(\frac{\bar{c}_{j+1}}{1 - \bar{c}_{j+1}} - \frac{\bar{c}_j}{1 - \bar{c}_j} \right) x_b^j,$$

and agent i's optimal consumption bundle becomes

$$x_g^i = (1 - p_i)e^i - \frac{p_i - \bar{c}_i}{1 - \bar{c}_i}$$
$$x_b^i = \frac{p_i - \bar{c}_i}{\bar{c}_i} + \frac{(1 - \bar{c}_i)p_i}{\bar{c}_i} e^i.$$

Given these setup, the parameters of the model are as in the following Table C.1.

Parameter	Description	Value
e	good state endowment	10
$(p_1, p_2, \dots, p_5)$	probability of bad state	$(0.1, 0.15, \dots, 0.3)$
$(m_1, m_2, \dots, m_5)$	mass of each type	$(0.2, 0.2, \dots, 0.2)$
γ	outside option utility level	0.0758

Table C.1: Parameter values for numerical simulations

For the numerical procedure, we check **Condition MS(i)** starting from $i = 1$ and updating the endowment of $e^{i+1} = e$ whenever the condition is satisfied. The algorithm repeats this until it finds the agent that does not violate the test. Then, we proceed with the rest of the agents' consumption quantities using the iterative representation.

The algorithm that solves this numerical model is the following: For each i,

1. Calculate the endowment of agent i, e^i, using the previous agents' quantities.
2. Derive the optimal quantity for agent i, (x_g^i, x_b^i) under e^i and $\bar{c}_i$.
3. Compute the utility level $v_i(x_g^i, x_b^i)$ and compare that to $v_i(e, 0) + \gamma$.
4. If it is above $v_i(e, 0) + \gamma$, then the (x_g^i, x_b^i) quantity is the optimal quantity. Otherwise, set $(x_g^i, x_b^i) = (e, 0)$.
5. Move to the next agent $i + 1$ and repeat from the first step.